

Project Data Page:





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Instrument Air reciever

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Instrument Air reciever-7%.pvdb

Date of Analysis : Apr 20,2026 12:06pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 5506.000 mm. ID
Vessel Design Length, Tangent to Tangent 13000.00 mm.
Distance of Bottom Tangent above Grade 3000.00 mm.
Distance of Base above Grade 300.00 mm.
Specified Datum Line Distance 2700.00 mm.
Internal Design Temperature 82 °C
Internal Design Pressure 10.987 bars
External Design Temperature 0 °C
Maximum Allowable Working Pressure 11.4 bars
Shop Test Pressure 15.780 bars
Field Test Pressure 14.283 bars
Required Minimum Design Metal Temperature -1.0 °C
Warmest Computed Minimum Design Metal Temperature -6.0 °C
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-516 70	...	...	K02700	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt2	...	...	...	1.0000	N/A	No
Skirt-1	...	...	...	1.0000	N/A	No
Bottom Dish	10.989	0.00	11.500	1.0000	No	No
Shell-1	10.988	0.00	11.400	1.0000	N/A	No
Shell2	10.988	0.00	11.400	1.0000	N/A	No
Shell3	10.988	0.00	11.400	1.0000	N/A	No
shell4	10.988	0.00	11.400	1.0000	N/A	No
shell5	10.987	0.00	11.400	1.0000	N/A	No
Top dish	10.987	0.00	11.500	1.0000	No	No

Liquid Level: 15753.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.001

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-1000.0	1700.0	22.0	22.0	...	...	0.70	0.70
Skirt	0.0	1000.0	18.0	18.0	...	...	0.70	0.50
Ellipse	50.0	50.0	28.0	24.0	22.9	...	1.00	1.00
Cylinder	3050.0	3000.0	28.0	28.0	26.9	...	0.85	0.85
Cylinder	6050.0	3000.0	28.0	28.0	26.9	...	0.85	0.85
Cylinder	9050.0	3000.0	28.0	28.0	26.9	...	0.85	0.85
Cylinder	11050.0	2000.0	28.0	28.0	26.9	...	0.85	0.85
Cylinder	12950.0	1900.0	28.0	28.0	26.9	...	0.85	0.85
Ellipse	13000.0	50.0	28.0	24.0	22.9	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	0.829	n/a	Passed
Trunnion	WRC-107/537	0.831	n/a	Passed
Trunnion	WRC-107/537	0.833	n/a	Passed
Trunnion	WRC-107/537	0.834	n/a	Passed
Trunnion	WRC-107/537	0.836	n/a	Passed
Trunnion	WRC-107/537	0.837	n/a	Passed
Trunnion	WRC-107/537	0.838	n/a	Passed
Trunnion	WRC-107/537	0.838	n/a	Passed
Trunnion	WRC-107/537	0.839	n/a	Passed
Trunnion	WRC-107/537	0.839	n/a	Passed
Trunnion	WRC-107/537	0.838	n/a	Passed
Trunnion	WRC-107/537	0.838	n/a	Passed
Trunnion	WRC-107/537	0.837	n/a	Passed
Trunnion	WRC-107/537	0.836	n/a	Passed
Trunnion	WRC-107/537	0.835	n/a	Passed
Trunnion	WRC-107/537	0.833	n/a	Passed
Trunnion	WRC-107/537	0.831	n/a	Passed
Trunnion	WRC-107/537	0.829	n/a	Passed
Trunnion	WRC-107/537	0.827	n/a	Passed
Trunnion	WRC-107/537	0.824	n/a	Passed
Trunnion	WRC-107/537	0.821	n/a	Passed
Trunnion	WRC-107/537	0.818	n/a	Passed
Trunnion	WRC-107/537	0.815	n/a	Passed
Trunnion	WRC-107/537	0.811	n/a	Passed
Trunnion	WRC-107/537	0.807	n/a	Passed
Trunnion	WRC-107/537	0.803	n/a	Passed
Trunnion	WRC-107/537	0.799	n/a	Passed
Trunnion	WRC-107/537	0.794	n/a	Passed
Trunnion	WRC-107/537	0.789	n/a	Passed
Trunnion	WRC-107/537	0.784	n/a	Passed
Trunnion	WRC-107/537	0.778	n/a	Passed
Trunnion	WRC-107/537	0.772	n/a	Passed
Trunnion	WRC-107/537	0.766	n/a	Passed
Trunnion	WRC-107/537	0.760	n/a	Passed
Trunnion	WRC-107/537	0.754	n/a	Passed
Trunnion	WRC-107/537	0.747	n/a	Passed
Trunnion	WRC-107/537	0.740	n/a	Passed
Trunnion	WRC-107/537	0.733	n/a	Passed
Trunnion	WRC-107/537	0.725	n/a	Passed
Trunnion	WRC-107/537	0.717	n/a	Passed
Trunnion	WRC-107/537	0.709	n/a	Passed
Trunnion	WRC-107/537	0.701	n/a	Passed
Trunnion	WRC-107/537	0.693	n/a	Passed
Trunnion	WRC-107/537	0.684	n/a	Passed
Trunnion	WRC-107/537	0.675	n/a	Passed
Trunnion	WRC-107/537	0.666	n/a	Passed
Trunnion	WRC-107/537	0.657	n/a	Passed
Trunnion	WRC-107/537	0.647	n/a	Passed
Trunnion	WRC-107/537	0.638	n/a	Passed
Trunnion	WRC-107/537	0.628	n/a	Passed

Trunnion	WRC-107/537	0.617	n/a	Passed
Trunnion	WRC-107/537	0.607	n/a	Passed
Trunnion	WRC-107/537	0.596	n/a	Passed
Trunnion	WRC-107/537	0.585	n/a	Passed
Trunnion	WRC-107/537	0.574	n/a	Passed
Trunnion	WRC-107/537	0.563	n/a	Passed
Trunnion	WRC-107/537	0.552	n/a	Passed
Trunnion	WRC-107/537	0.540	n/a	Passed
Trunnion	WRC-107/537	0.528	n/a	Passed
Trunnion	WRC-107/537	0.516	n/a	Passed
Trunnion	WRC-107/537	0.504	n/a	Passed
Trunnion	WRC-107/537	0.491	n/a	Passed
Trunnion	WRC-107/537	0.479	n/a	Passed
Trunnion	WRC-107/537	0.466	n/a	Passed
Trunnion	WRC-107/537	0.453	n/a	Passed
Trunnion	WRC-107/537	0.445	n/a	Passed
Trunnion	WRC-107/537	0.451	n/a	Passed
Trunnion	WRC-107/537	0.457	n/a	Passed
Trunnion	WRC-107/537	0.463	n/a	Passed
Trunnion	WRC-107/537	0.469	n/a	Passed
Trunnion	WRC-107/537	0.475	n/a	Passed
Trunnion	WRC-107/537	0.482	n/a	Passed
Trunnion	WRC-107/537	0.488	n/a	Passed
Trunnion	WRC-107/537	0.495	n/a	Passed
Trunnion	WRC-107/537	0.502	n/a	Passed
Trunnion	WRC-107/537	0.509	n/a	Passed
Trunnion	WRC-107/537	0.516	n/a	Passed
Trunnion	WRC-107/537	0.524	n/a	Passed
Trunnion	WRC-107/537	0.532	n/a	Passed
Trunnion	WRC-107/537	0.541	n/a	Passed
Trunnion	WRC-107/537	0.550	n/a	Passed
Trunnion	WRC-107/537	0.559	n/a	Passed
Trunnion	WRC-107/537	0.569	n/a	Passed
Trunnion	WRC-107/537	0.580	n/a	Passed
Trunnion	WRC-107/537	0.592	n/a	Passed
Trunnion	WRC-107/537	0.605	n/a	Passed
Trunnion	WRC-107/537	0.620	n/a	Passed
Trunnion	WRC-107/537	0.636	n/a	Passed
Trunnion	WRC-107/537	0.654	n/a	Passed
Trunnion	WRC-107/537	0.675	n/a	Passed
Trunnion	WRC-107/537	0.699	n/a	Passed
Trunnion	WRC-107/537	0.839	n/a	Passed
N2-3"	WRC-107/537	0.893	n/a	Passed
N6-2"	WRC-297	0.848	Shell	Passed
N1-6"	WRC-297	0.848	Shell	Passed
N3-6"	WRC-107/537	0.893	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	5947.16	19251.8	54602.7	231765
20	30	2200	5503.62	19105.1	44869.4	199161
30	40	2725	5242.9	18964.9	39496	180125
40	50	4250	4974.19	18155.4	39240.6	179197
50	60	7250	4190.76	17090.6	25492.9	126327
60	70	10250	3166.61	14845.4	14456.6	78422.1
70	80	12750	2113.17	11553.5	6536.77	38822.9
80	90	14700	1590.88	9427.53	2832.65	17841.4
90	100	15675	842.928	5668.34	520.485	3500.04

Abs Max of the all of the Stress Ratio's : 0.4727

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing

38.0000 mm.



Inside Diameter of Basing	5389.0000	mm.
Outside Diameter of Basing	5989.0000	mm.
Nominal Diameter of Bolts	64.0000	mm.
Diameter of Bolt Circle	5773.0000	mm.
Number of Bolts	24	
Thickness of Gusset Plates	22.0000	mm.
Average Width of Gusset Plates	206.0000	mm.
Height of Gussets	312.0000	mm.
Distance between Gussets	120.0000	mm.
Thickness of Top Plate or Ring	50.0000	mm.
Radial Width of the Top Plate	216.0000	mm.

Basing Sketch for Cylindrical Skirts:

Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	5947.	Kgf
Total Earthquake Shear on Support	19252.	Kgf
Wind Moment on Support	61492.	Kg-m.
Earthquake Moment on Support	238654.	Kg-m.
Maximum Initial Bolt Load	2125.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	92179.3	kg.
Shop Test - Fabricated + water (full)	448258.6	kg.
Shipping - Fab. + rem. intls.+ shipping app.	92179.3	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	117705.0	kg.
Empty - Fab. + intls. + details + wghts.	117705.0	kg.
Operating - Empty + operating liquid (no ca)	118057.3	kg.
Field Test - Empty weight + test medium (full)	469892.9	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2-3"	11.5	...	...	OK	5.80	OK	Passed
K1B-2"	11.4	...	...	OK	5.80	OK	Passed
N6-2"	11.4	...	...	OK	5.80	OK	Passed
K1A-2"	11.4	...	...	OK	5.80	OK	Passed
M1-24"	11.4	...	...	...		OK	Passed
N1-6"	11.4	...	...	OK	7.22	OK	Passed
N3-6"	11.5	...	...	OK	7.22	OK	Passed
N4-3"	11.5	...	...	OK	5.80	OK	Passed
N5-2"	11.5	...	...	OK	5.80	OK	Passed
K3-2"	11.5	...	...	OK	5.80	OK	Passed
K2-2"	11.5	...	...	OK	5.80	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 11.4 Nozzle : N6-2"
 Minimum MAWP Shells/Flanges : 11.4 Element : Shell-1
 Minimum MAPnc Shells/Flanges : 11.9 Element : shell15

Computed Vessel M.A.W.P. : 11.4 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	N2-3"	0.000	0.000	142.862
40	K1B-2"	1550.000	270.000	138.328
40	N6-2"	1500.000	90.000	138.328
50	K1A-2"	4550.000	270.000	138.328
60	M1-24"	7550.000	270.000	1119.200
80	N1-6"	12000.000	270.000	290.588
90	N3-6"	0.000	0.000	290.588
90	N4-3"	0.000	90.000	142.862
90	N5-2"	0.000	120.000	130.328
90	K3-2"	0.000	270.000	130.328
90	K2-2"	0.000	250.000	130.328

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1B-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.32	300
N6-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	234.70	300
K1A-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.32	300
N5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	439.87	300
K3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	437.57	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	439.87	300
N2-3"	3.000 in	160	WNF	3.500	11.125	190.00	28.00	224.20	300
N4-3"	3.000 in	160	WNF	3.500	11.125	188.90	28.00	439.45	150
N1-6"	6.000 in	120	WNF	6.625	14.275	270.00	26.00	229.29	300
N3-6"	6.000 in	120	WNF	6.625	14.275	270.00	28.00	424.71	300
M1-24"	24.000 in	Actual	WNF	24.000	26.000	1009.60	28.00	194.93	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1B-2"	SA-105	24.000	10.000	...	...	...
N6-2"	SA-105	24.000	10.000	...	...	...
K1A-2"	SA-105	24.000	10.000	...	...	...
N5-2"	SA-105	24.000	10.000	...	...	...
K3-2"	SA-105	24.000	10.000	...	...	...
K2-2"	SA-105	24.000	10.000	...	...	...
N2-3"	SA-106 B	24.000	10.000	20.000	28.000	...
N4-3"	SA-106 B	24.000	10.000	18.000	28.000	...
N1-6"	SA-106 B	26.000	10.000	20.000	26.000	...
N3-6"	SA-106 B	24.000	10.000	18.000	28.000	...
M1-24"	SA-516 70	26.000	10.000	20.000	26.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K1B-2"	1550.000	270.0	200.00	0.00	Shell-1
N6-2"	1500.000	90.0	206.38	0.00	Shell-1
K1A-2"	4550.000	270.0	200.00	0.00	Shell12
N5-2"	...	120.0	400.00	0.00	Top dish
K3-2"	...	270.0	400.00	0.00	Top dish
K2-2"	...	250.0	400.00	0.00	Top dish
N2-3"	...	0.0	200.00	0.00	Bottom Dish
N4-3"	...	90.0	400.00	0.00	Top dish
N1-6"	12000.000	270.0	200.00	0.00	shell15
N3-6"	...	0.0	400.00	0.00	Top dish
M1-24"	7550.000	270.0	150.00	0.00	Shell13



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 28.0mm. THK X 5506.0mm. ID X 50.0mm.	SA-516 70
4	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 2700.0mm	...
2	CLADDING: 1426.500mm X 3.000mm THK	...
3	CLADDING: 3000.000mm X 3.000mm THK	...
3	LADDER: 4000.0mm	...
1	CLADDING: 2000.000mm X 3.000mm THK	...
1	CLADDING: 1900.000mm X 3.000mm THK	...
2	TRUNNIONS:	SA-516 70
2	TRUNNION PADS: 900.0mm OD X 550.0mm ID X 28.0mm THK	
1	PLATFORM: 1000mm X 6000mm WIDE	
1	LADDER: 2820.0mm	...
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 5989.0mm. OD X 5389.0 ID X 38.0mm.	SA-516 70
24	BASERING BOLTS: 64.0mm. DIA	SA-307 A
24	BASERING BOLT NUTS: 64.0mm. DIA	...
1	BASERING TOP RING: 5989.0mm. OD X 5557.0 ID X 50.0mm.	...
48	BASERING GUSSET PLATES: 206.0mm. X 206.0mm. X 312.0mm. X 22.0mm.	...
5	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 3051.8mm. X 1700.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 2060.2mm. X 1700.0mm.	SA-36
5	CYLINDRICAL SKIRT SEGMENT 18.0mm. THK X 3054.0mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm. THK X 2061.7mm. X 1000.0mm.	SA-516 70
13	CYLINDRICAL SEGMENT 28.0mm. THK X 3048.0mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm. THK X 2057.6mm. X 3000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 28.0mm. THK X 14249.4mm. X 3000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 28.0mm. THK X 9143.9mm. X 3000.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 28.0mm. THK X 3048.0mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm. THK X 14249.4mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm. THK X 9143.9mm. X 2000.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 28.0mm. THK X 3048.0mm. X 1900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm. THK X 14249.4mm. X 1900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm. THK X 9143.9mm. X 1900.0mm.	SA-516 70

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Bill of Material:



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Step: 40 12:06pm Apr 20,2026



HEXAGON

1

NAMEPLATE

...

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom Dish	[10]	B	-2	-3	-29	0.992	24.000	1.00	No
Bottom Dish	[7]	B	2	-6		0.849	28.000	1.00	No
Shell-1	[8]	B	2	-6		0.850	28.000	0.85	No
Shell2	[8]	B	2	-6		0.850	28.000	0.85	No
Shell3	[8]	B	2	-6		0.850	28.000	0.85	No
shell4	[8]	B	2	-6		0.850	28.000	0.85	No
shell5	[8]	B	2	-6		0.850	28.000	0.85	No
Top dish	[10]	B	-2	-3	-29	0.992	24.000	1.00	No
Top dish	[7]	B	2	-6		0.849	28.000	1.00	No
N2-3"	[1]	B	-2	-29	-29	0.893	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1B-2"	[1]	A	7	-104		0.014	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-2"	[1]	A	7	-104		0.014	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A-2"	[1]	A	7	-104		0.014	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-24"	[1]	B	2	-6		0.849	28.000	1.00	No
Nozzle Flg	[5]	A	-18	-42					
N1-6"	[1]	B	0	-8		0.849	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-6"	[1]	B	-2	-29	-29	0.893	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4-3"	[1]	B	-2	-29	-29	0.893	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-42					
N5-2"	[1]	A	7	-104		0.014	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3-2"	[1]	A	7	-104		0.014	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2-2"	[1]	A	7	-104		0.014	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 7 -3

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -6.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Instrument Air reciever

Design Internal Pressure (for Hydrotest)	10.987	bars
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	500	mm.
Minimum Design Metal Temperature	-1.0	°C
Type of Construction	Welded	
Special Service	Air/Water/Steam	
Degree of Radiography	RT-2	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	User Entered Pressure
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.020
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes



Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt2
Distance "FROM" to "TO"	1700 mm.
Skirt Outside Diameter	5557 mm.
Diameter of Skirt at Base	5557 mm.
Skirt Thickness	22 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-36
Allowable Stress, Ambient	114 N./mm ²
Allowable Stress, Operating	114 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02600
Product Form	Plate; sheet
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/9
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	5553 mm.
Diameter of Skirt at Base	5553 mm.
Skirt Thickness	18 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	18 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element From Node	20
Detail Type	Weight
Detail ID	Skirt beam
Dist. from "FROM" Node / Offset dist	-0.0001463 mm.
Miscellaneous Weight	825 Kg



Offset from Element Centerline 0 mm.

Element# 3/9
Element From Node 30
Element To Node 40
Element Type Elliptical
Description Bottom Dish
Distance "FROM" to "TO" 50 mm.
Inside Diameter 5506 mm.
Element Thickness 24 mm.
Internal Corrosion Allowance 1 mm.
Nominal Thickness 28 mm.
External Corrosion Allowance 0 mm.
Design Internal Pressure 10.987 bars
Design Temperature Internal Pressure 82 °C
Design External Pressure 0 bars
Design Temperature External Pressure 0 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Elliptical Head Factor 2.0
Weld is pre-Heated No

Element From Node 30
Detail Type Platform
Detail ID PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist 0 mm.
Platform Start Angle (degrees) 360.0
Platform End Angle (degrees) 0.0
Platform Wind Area 30000 cm²
Platform Weight 5890.1 Kg
Platform Railing Weight 0 Kg/mm.
Platform Grating Weight 0.017 Kgs/cm²
Platform Width 1500 mm.
Platform Height 1000 mm.
Platform Clearance or End Offset 85 mm.
Platform Force Coefficient 1.0
Ladder Layout Angle 90.0
Ladder Start Elevation 0 mm.
Ladder End Elevation 2700 mm.
Unit Weight of Ladder 0.037 Kg/mm.
Platform Length (top head platform) 0 mm.

Element From Node 30
Detail Type Liquid
Detail ID Liquid 30
Dist. from "FROM" Node / Offset dist -1376.5 mm.
Height/Length of Liquid 1426.5 mm.
Liquid Density 0.99840E-06 kg./cm³

Element From Node 30
Detail Type Lining
Detail ID clad
Dist. from "FROM" Node / Offset dist -1376.5 mm.
Height/Length of Lining 1426.5 mm.
Thickness of Lining 3 mm.
Density 0.008 kg./cm³

Element From Node 30
Detail Type Nozzle
Detail ID N2-3"
Dist. from "FROM" Node / Offset dist 0 mm.
Nozzle Diameter 3 in.
Nozzle Schedule 160
Nozzle Class 300
Layout Angle 0.0



Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	4/9
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	5506 mm.
Element Thickness	28 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	28 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10.987 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.99840E-06 kg./cm³

Element From Node	40
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	3000 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm³

Element From Node	40
Detail Type	Nozzle
Detail ID	K1B-2"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	40
Detail Type	Nozzle
Detail ID	N6-2"
Dist. from "FROM" Node / Offset dist	1450 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105



Element#	5/9
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell12
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	5506 mm.
Element Thickness	28 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	28 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10.987 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	50
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	1000 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	180.0
Platform Wind Area	30000 cm²
Platform Weight	2978.2 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	0 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	4000 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.99840E-06 kg./cm³

Element From Node	50
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	3000 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm³

Element From Node	50
Detail Type	Nozzle
Detail ID	K1A-2"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1



Nozzle Matl

SA-105

Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell3
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	5506 mm.
Element Thickness	28 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	28 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10.987 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	60
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	2000 mm.
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	30000 cm²
Platform Weight	2978.2 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	0 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	4000 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.99840E-06 kg./cm³

Element From Node	60
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	3000 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm³

Element From Node	60
Detail Type	Nozzle
Detail ID	M1-24"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	Y



Weight of Nozzle (Used if > 0) 0 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-516 70

Element From Node 60
Detail Type Weight
Detail ID Davit-1
Dist. from "FROM" Node / Offset dist 1500 mm.
Miscellaneous Weight 50 Kgf
Offset from Element Centerline 3800 mm.

Element From Node 60
Detail Type Weight
Detail ID Nozzle Clad
Dist. from "FROM" Node / Offset dist 0 mm.
Miscellaneous Weight 50 Kgf
Offset from Element Centerline 0 mm.

Element# 7/9
Element From Node 70
Element To Node 80
Element Type Cylinder
Description shell4
Distance "FROM" to "TO" 2000 mm.
Inside Diameter 5506 mm.
Element Thickness 28 mm.
Internal Corrosion Allowance 1 mm.
Nominal Thickness 28 mm.
External Corrosion Allowance 0 mm.
Design Internal Pressure 10.987 bars
Design Temperature Internal Pressure 82 °C
Design External Pressure 0 bars
Design Temperature External Pressure 0 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70
Efficiency, Longitudinal Seam 0.85
Efficiency, Circumferential Seam 0.85
Weld is pre-Heated No

Element From Node 70
Detail Type Liquid
Detail ID Liquid 70
Dist. from "FROM" Node / Offset dist 0 mm.
Height/Length of Liquid 2000 mm.
Liquid Density 0.99840E-06 kg./cm³

Element From Node 70
Detail Type Lining
Detail ID clad
Dist. from "FROM" Node / Offset dist 0 mm.
Height/Length of Lining 2000 mm.
Thickness of Lining 3 mm.
Density 0.008 kg./cm³

Element# 8/9
Element From Node 80
Element To Node 90
Element Type Cylinder
Description shell5
Distance "FROM" to "TO" 1900 mm.
Inside Diameter 5506 mm.
Element Thickness 28 mm.
Internal Corrosion Allowance 1 mm.
Nominal Thickness 28 mm.
External Corrosion Allowance 0 mm.

Design Internal Pressure	10.987	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	600	mm.
Platform Start Angle (degrees)	360.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	2978.2	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	0	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	4000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid 80	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1900	mm.
Liquid Density	0.99840E-06	kg./cm ³

Element From Node	80	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	1900	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm ³

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N1-6"	
Dist. from "FROM" Node / Offset dist	950	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Weight	
Detail ID	Clip wt	
Dist. from "FROM" Node / Offset dist	0.001161	mm.
Miscellaneous Weight	1450	Kgf
Offset from Element Centerline	0	mm.



Element To Node	100
Element Type	Elliptical
Description	Top dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	5506 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	28 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10.987 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Platform
Detail ID	SQR PTL
Dist. from "FROM" Node / Offset dist	1500 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	60000 cm²
Platform Weight	6227.5 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm²
Platform Width	6000 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	3000 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	180 mm.
Ladder End Elevation	3000 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	6000 mm.

Element From Node	90
Detail Type	Liquid
Detail ID	Liquid 90
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1426.5 mm.
Liquid Density	0.99840E-06 kg./cm³

Element From Node	90
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	1426.5 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm³

Element From Node	90
Detail Type	Nozzle
Detail ID	N3-6"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	90
-------------------	----



Detail Type	Nozzle
Detail ID	N4-3"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	90
Detail Type	Nozzle
Detail ID	N5-2"
Dist. from "FROM" Node / Offset dist	1600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	120.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	90
Detail Type	Nozzle
Detail ID	K3-2"
Dist. from "FROM" Node / Offset dist	1400 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	90
Detail Type	Nozzle
Detail ID	K2-2"
Dist. from "FROM" Node / Offset dist	1600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	250.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	90
Detail Type	Weight
Detail ID	PSV1-3"
Dist. from "FROM" Node / Offset dist	1400 mm.
Miscellaneous Weight	1500 Kg
Offset from Element Centerline	3000 mm.

Element From Node	90
Detail Type	Weight
Detail ID	PSV2-3"
Dist. from "FROM" Node / Offset dist	1400 mm.
Miscellaneous Weight	1500 Kg
Offset from Element Centerline	3000 mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Skirt2	...	1700	1700	...	1700
Skirt-1	...	2700	1000	...	1000
Bottom Dish	...	2750	50	...	50
Shell-1	...	5750	3000	...	3000
Shell2	...	8750	3000	...	3000
Shell3	...	11750	3000	...	3000
shell4	...	13750	2000	...	2000
shell5	...	15650	1900	...	1900
Top dish	...	15700	50	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt2	...	...	22	1	5557	...	1
Skirt-1	...	...	18	1	5553	...	2
Bottom Dish	10.989	10.989	28	1	5506	138	3
Shell-1	10.988	10.988	28	1	5506	117.3	4
Shell2	10.988	10.988	28	1	5506	117.3	5
Shell3	10.988	10.988	28	1	5506	117.3	6
shell4	10.988	10.988	28	1	5506	117.3	7
shell5	10.987	10.987	28	1	5506	117.3	8
Top dish	10.987	10.987	28	1	5506	138	9

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt2	...	...	No Calc	No Calc	22	No Calc	1
Skirt-1	...	...	No Calc	No Calc	18	No Calc	2
Bottom Dish	10.987	10.987	11.5	12	24	22.9378	3
Shell-1	10.987	10.987	11.4	11.9	28	26.9467	4
Shell2	10.987	10.987	11.4	11.9	28	26.946	5
Shell3	10.987	10.987	11.4	11.9	28	26.9453	6
shell4	10.987	10.987	11.4	11.9	28	26.9446	7
shell5	10.987	10.987	11.4	11.9	28	26.9441	8
Top dish	10.987	10.987	11.5	12	24	22.935	9

Minimum 11.4 11.9

MAWP: 11.4 bars, limited by: Shell-1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Bottom Dish

Elliptical Head Design Information:					
Design Pressure	P	10.989 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5506.000 mm.	Actual Stress		131.6 N./mm ²
Spec. Min. Thk t _e , or s _a		24.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		344076.8 cm ²	SG of Contents		0.001
Element Volume		23044424.0 cm ³	Weight of Contents		23.0 Kg _f
Empty Weight		8143.0 Kg _f	Operating Weight		8166.0 Kg _f
			Comp. All. Ope. Stress		103.6 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.825 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.413 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (11 \cdot 5508 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 11)$$

$$= 21.9378 + 1.0000 = 22.9378 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.002 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 23) / (1 \cdot 5508 + 0.2 \cdot 23)$$

$$= 11.5 - 0.0015 = 11.5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 24) / (1 \cdot 5506 + 0.2 \cdot 24)$$

$$= 12 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (11 \cdot (1 \cdot 5508 + 0.2 \cdot 23)) / (2 \cdot 1 \cdot 23)$$

$$= 131.630 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (11 \cdot 2754) / (138 \cdot 1 - 0.6 \cdot 11) + 1$$

$$= 23.036 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.001 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 27) / (2754 + 0.6 \cdot 27)$$

$$= 13.4 - 0.0014 = 13.4 \text{ bars}$$

Factor K, corroded condition [Kcor]:



$$\begin{aligned} &= (2 + (\text{Inside Diameter}/(2 \cdot \text{Inside Head Depth}))^2)/6 \\ &= (2 + (5508/(2 \cdot 1378))^2)/6 \\ &= 0.999516 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.210 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 24$, $t_r = 22.8$, $c = 1$ mm., $E^* = 1$
Thickness Ratio $= t_r \cdot E^*/(t_g - c) = 0.99$, Temp. Reduction $= 0$ °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-3 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 28$, $t_r = 22.9$, $c = 1$ mm., $E^* = 1$
Thickness Ratio $= t_r \cdot E^*/(t_g - c) = 0.85$, Temp. Reduction $= 8$ °C

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-6 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	11.432 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	28.000 mm.	Basic MDMT	2 °C
Stress/Thk. Ratio	0.849	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-6 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.825 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.413 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	10.988 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5506.000 mm.	Actual Stress		132.6 N./mm ²
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		524206.2 cm ²	SG of Contents		0.001
Element Volume		71443272.0 cm ³	Weight of Contents		71.2 Kg
Empty Weight		11318.8 Kg	Operating Weight		11390.1 Kg
			Comp. All. Ope. Stress		109.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (11 \cdot 2754) / (138 \cdot 0.85 - 0.6 \cdot 11)$$

$$= 25.9467 + 1.0000 = 26.9467 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 27) / (2754 + 0.6 \cdot 27)$$

$$= 11.4 - 0.0014 = 11.4 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 28) / (2753 + 0.6 \cdot 28)$$

$$= 11.9 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (11 \cdot (2754 + 0.6 \cdot 27)) / (0.85 \cdot 27)$$

$$= 132.644 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.506 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 28, t_r = 27, c = 1 mm., E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	11.432 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	28.000 mm.	Basic MDMT	2 °C

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Element: Shell-1



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Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-6 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell2

Cylindrical Shell Design Information:					
Design Pressure	P	10.988 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5506.000 mm.	Actual Stress		132.6 N./mm ²
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		524206.2 cm ²	SG of Contents		0.001
Element Volume		71443272.0 cm ³	Weight of Contents		71.2 Kg
Empty Weight		11318.8 Kg	Operating Weight		11390.1 Kg
			Comp. All. Ope. Stress		109.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (11 \cdot 2754) / (138 \cdot 0.85 - 0.6 \cdot 11)$$

$$= 25.9460 + 1.0000 = 26.9460 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 27) / (2754 + 0.6 \cdot 27)$$

$$= 11.4 - 0.0011 = 11.4 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 28) / (2753 + 0.6 \cdot 28)$$

$$= 11.9 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (11 \cdot (2754 + 0.6 \cdot 27)) / (0.85 \cdot 27)$$

$$= 132.641 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.506 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 28, t_r = 27, c = 1 mm., E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	11.432 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	28.000 mm.	Basic MDMT	2 °C

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Element: Shell2



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Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-6 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell3

Cylindrical Shell Design Information:					
Design Pressure	P	10.988 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5506.000 mm.	Actual Stress		132.6 N./mm ²
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		524206.2 cm ²	SG of Contents		0.001
Element Volume		71443272.0 cm ³	Weight of Contents		71.2 Kg
Empty Weight		11318.8 Kg	Operating Weight		11390.1 Kg
			Comp. All. Ope. Stress		109.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (11 \cdot 2754) / (138 \cdot 0.85 - 0.6 \cdot 11)$$

$$= 25.9453 + 1.0000 = 26.9453 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 27) / (2754 + 0.6 \cdot 27)$$

$$= 11.4 - 0.00082 = 11.4 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 28) / (2753 + 0.6 \cdot 28)$$

$$= 11.9 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (11 \cdot (2754 + 0.6 \cdot 27)) / (0.85 \cdot 27)$$

$$= 132.637 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.506 %

Minimum Design Metal Temperature Results:

Govern. thk, t_g = 28, t_r = 27, c = 1 mm., E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	11.431 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	28.000 mm.	Basic MDMT	2 °C

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Element: Shell3



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Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-6 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell4

Cylindrical Shell Design Information:					
Design Pressure	P	10.988 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5506.000 mm.	Actual Stress		132.6 N./mm ²
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		349470.8 cm ²	SG of Contents		0.001
Element Volume		47628852.0 cm ³	Weight of Contents		47.5 Kg _f
Empty Weight		7545.9 Kg _f	Operating Weight		7593.4 Kg _f
			Comp. All. Ope. Stress		109.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (11 \cdot 2754) / (138 \cdot 0.85 - 0.6 \cdot 11)$$

$$= 25.9446 + 1.0000 = 26.9446 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 27) / (2754 + 0.6 \cdot 27)$$

$$= 11.4 - 0.00052 = 11.4 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 28) / (2753 + 0.6 \cdot 28)$$

$$= 11.9 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (11 \cdot (2754 + 0.6 \cdot 27)) / (0.85 \cdot 27)$$

$$= 132.634 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.506 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 28, tr = 27, c = 1 mm., E* = 0.85

Thickness Ratio = tr * E* / (tg - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	11.431 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	28.000 mm.	Basic MDMT	2 °C

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Element: shell4



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Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-6 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: shell15

Cylindrical Shell Design Information:					
Design Pressure	P	10.987 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5506.000 mm.	Actual Stress		132.6 N./mm ²
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		331997.2 cm ²	SG of Contents		0.001
Element Volume		45247404.0 cm ³	Weight of Contents		45.1 Kg
Empty Weight		7168.6 Kg	Operating Weight		7213.7 Kg
			Comp. All. Ope. Stress		109.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (11 \cdot 2754) / (138 \cdot 0.85 - 0.6 \cdot 11)$$

$$= 25.9441 + 1.0000 = 26.9441 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 27) / (2754 + 0.6 \cdot 27)$$

$$= 11.4 - 0.00033 = 11.4 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 28) / (2753 + 0.6 \cdot 28)$$

$$= 11.9 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (11 \cdot (2754 + 0.6 \cdot 27)) / (0.85 \cdot 27)$$

$$= 132.631 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.506 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 28, t_r = 27, c = 1 mm., E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	11.431 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	28.000 mm.	Basic MDMT	2 °C

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Element: shell5



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Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-6 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Top dish

Elliptical Head Design Information:					
Design Pressure	P	10.987 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5506.000 mm.	Actual Stress		131.6 N./mm ²
Spec. Min. Thk t _e , or s _a		24.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		344076.8 cm ²	SG of Contents		0.001
Element Volume		23044424.0 cm ³	Weight of Contents		23.0 Kg _f
Empty Weight		8143.0 Kg _f	Operating Weight		8166.0 Kg _f
			Comp. All. Ope. Stress		103.6 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.825 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.413 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (11 \cdot 5508 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 11)$$

$$= 21.9350 + 1.0000 = 22.9350 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 23) / (1 \cdot 5508 + 0.2 \cdot 23)$$

$$= 11.5 - 0.00013 = 11.5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 24) / (1 \cdot 5506 + 0.2 \cdot 24)$$

$$= 12 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (11 \cdot (1 \cdot 5508 + 0.2 \cdot 23)) / (2 \cdot 1 \cdot 23)$$

$$= 131.613 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (11 \cdot 2754) / (138 \cdot 1 - 0.6 \cdot 11) + 1$$

$$= 23.033 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 27) / (2754 + 0.6 \cdot 27)$$

$$= 13.4 - 0.00013 = 13.4 \text{ bars}$$

Factor K, corroded condition [Kcor]:



$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (5508 / (2 \cdot 1378))^2) / 6$$

$$= 0.999516$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.210 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 24$, $t_r = 22.8$, $c = 1$ mm., $E^* = 1$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction $= 0$ °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -3 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 28$, $t_r = 22.9$, $c = 1$ mm., $E^* = 1$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction $= 8$ °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	11.431 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	28.000 mm.	Basic MDMT	2 °C
Stress/Thk. Ratio	0.849	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-6 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.825 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.413 mm.

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Hydrostatic Test Pressure Results:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 14.860 bars
UG99b(b)(35) = 1.30 * Design Pres * Sa/S 14.283 bars
UG99(c) = 1.30 * M.A.P. - Head(Hyd) 14.776 bars
UG100 = 1.10 * M.A.W.P. * Sa/S 12.574 bars
PED = max(1.43*DP, 1.25*DP*ratio) 15.711 bars
PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 15.711 bars
App 27-4 = M.A.W.P. 11.430 bars

User Defined Hydrostatic Test Pressure at High Point 15.780 bars

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	187.4	234.0	0.801	16.32
Shell-1	189.9	234.0	0.812	16.32
Shell2	189.9	234.0	0.812	16.32
Shell3	189.9	234.0	0.812	16.32
shell4	189.9	234.0	0.812	16.32
shell5	189.9	234.0	0.812	16.32
Top dish	187.4	234.0	0.801	16.32

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-3"	Nozzle	118.00	118.00	1.000
N2-3"	Pad	138.00	138.00	1.000
K1B-2"	Nozzle	138.00	138.00	1.000
N6-2"	Nozzle	138.00	138.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
N1-6"	Nozzle	118.00	118.00	1.000
N1-6"	Pad	138.00	138.00	1.000
N3-6"	Nozzle	118.00	118.00	1.000
N3-6"	Pad	138.00	138.00	1.000
N4-3"	Nozzle	118.00	118.00	1.000
N4-3"	Pad	138.00	138.00	1.000
N5-2"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell2	138.00	138.00	1.000
Shell3	138.00	138.00	1.000
shell4	138.00	138.00	1.000
shell5	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-3"	6.80	153.40	0.044
K1B-2"	3.45	179.40	0.019
N6-2"	3.45	179.40	0.019
K1A-2"	3.45	179.40	0.019
M1-24"	18.48	179.40	0.103
N1-6"	10.34	153.40	0.067
N3-6"	10.34	153.40	0.067
N4-3"	6.80	153.40	0.044
N5-2"	3.45	179.40	0.019
K3-2"	3.45	179.40	0.019
K2-2"	3.45	179.40	0.019

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	14.860 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	14.283 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	15.039 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	12.574 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	15.711 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	15.711 bars
App 27-4	= M.A.W.P.	11.430 bars

UG-99(b) Note 35, Test Pressure Calculation:
= Test Factor • Design Pressure • Stress Ratio
= 1.3 • 11 • 1
= 14.283 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	190.2	234.0	0.813	15.88
Shell-1	190.0	234.0	0.812	15.74
Shell12	186.4	234.0	0.797	15.44
Shell13	182.9	234.0	0.781	15.15
shell14	179.3	234.0	0.766	14.85
shell15	176.9	234.0	0.756	14.66
Top dish	173.4	234.0	0.741	14.47

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-3"	Nozzle	118.00	118.00	1.000
N2-3"	Pad	138.00	138.00	1.000
K1B-2"	Nozzle	138.00	138.00	1.000
N6-2"	Nozzle	138.00	138.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
N1-6"	Nozzle	118.00	118.00	1.000
N1-6"	Pad	138.00	138.00	1.000
N3-6"	Nozzle	118.00	118.00	1.000
N3-6"	Pad	138.00	138.00	1.000
N4-3"	Nozzle	118.00	118.00	1.000
N4-3"	Pad	138.00	138.00	1.000
N5-2"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell12	138.00	138.00	1.000
Shell13	138.00	138.00	1.000
shell14	138.00	138.00	1.000
shell15	138.00	138.00	1.000
Top dish	138.00	138.00	1.000



Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-3"	6.61	153.40	0.043
K1B-2"	3.33	179.40	0.019
N6-2"	3.33	179.40	0.019
K1A-2"	3.27	179.40	0.018
M1-24"	17.15	179.40	0.096
N1-6"	9.29	153.40	0.061
N3-6"	9.17	153.40	0.060
N4-3"	6.03	153.40	0.039
N5-2"	3.06	179.40	0.017
K3-2"	3.06	179.40	0.017
K2-2"	3.06	179.40	0.017

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	2700	...	...	No Calc	No Calc
20	30	2700	...	...	No Calc	No Calc
30	40	No Calc	5554	23	0.00057516	57.5032
40	50	13917.7	5562	27	0.00017271	17.2668
50	60	13917.7	5562	27	0.00017271	17.2668
60	70	13917.7	5562	27	0.00017271	17.2668
70	80	13917.7	5562	27	0.00017271	17.2668
80	90	13917.7	5562	27	0.00017271	17.2668
90	100	No Calc	5554	23	0.00057516	57.5032

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	24	No Calc	...	2.65
40	50	28	No Calc	...	1.12
50	60	28	No Calc	...	1.12
60	70	28	No Calc	...	1.12
70	80	28	No Calc	...	1.12
80	90	28	No Calc	...	1.12
90	100	24	No Calc	...	2.65

Minimum

1.12

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	2700	No Calc	No Calc	No Calc
20	30	2700	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	13917.7	No Calc	No Calc	No Calc
50	60	13917.7	No Calc	No Calc	No Calc
60	70	13917.7	No Calc	No Calc	No Calc
70	80	13917.7	No Calc	No Calc	No Calc
80	90	13917.7	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm³	Corroded Metal Wgt. kg.	Corroded ID Volume cm³	Extra due Misc % kg.
10	20	8663.76	...	8435.5	...	606.464
20	30	2425.87	...	2291.53	...	169.811
30	40	8143.04	23044424	7852.22	23069110	570.013
40	50	11318.8	71443272	10916.5	71495184	792.318
50	60	11318.8	71443272	10916.5	71495184	792.318
60	70	11318.8	71443272	10916.5	71495184	792.318
70	80	7545.86	47628852	7277.69	47663460	528.21
80	90	7168.55	45247404	6913.78	45280284	501.799
90	100	8143.04	23044424	7852.22	23069110	570.013
Total		76046	0.353E+09	73372	0.354E+09	5323

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
20	Wght	825	...	-0.0001463	...	Skirt beam
30	Plat	6302.37	...	...	0.32234E-04	PLAT:[1 OF 1]
30	Liqd	22.9564	...	-688.25	...	Liquid 30
30	Lini	891.652	...	-663.25	...	clad
30	Nozl	18.3891	...	-1514.5	...	N2-3"
40	Liqd	71.2441	...	1500	...	Liquid 40
40	Lini	1246.28	...	1500	...	Clad
40	Nozl	11.3058	-2853	1500	...	K1B-2"
40	Nozl	11.4939	2856.19	1450	...	N6-2"
50	Plat	3186.63	-2264.15	1000	...	PLAT:[1 OF 1]
50	Liqd	71.2441	...	1500	...	Liquid 50
50	Lini	1246.28	...	1500	...	clad
50	Nozl	11.3058	-2853	1500	...	K1A-2"
60	Plat	3186.63	2264.15	2000	...	PLAT:[1 OF 1]
60	Liqd	71.2441	...	1500	...	Liquid 60
60	Lini	1246.28	...	1500	...	clad
60	Nozl	527.517	-2828.01	1500	...	M1-24"
60	Wght	50	-3800	1500	...	Davit-1
60	Wght	50	...	...	...	Nozzle Clad
70	Liqd	47.4961	...	1000	...	Liquid 70
70	Lini	830.852	...	1000	...	clad
80	Plat	3186.63	-2264.15	600	...	PLAT:[1 OF 1]
80	Liqd	45.1213	...	950	...	Liquid 80
80	Lini	789.309	...	950	...	clad
80	Nozl	42.4082	-2853	950	...	N1-6"
80	Wght	1450	...	0.0011613	...	Clip wt
90	Plat	6663.42	...	1500	...	SQR PTL
90	Liqd	22.9564	...	738.25	...	Liquid 90
90	Lini	891.652	...	713.25	...	clad
90	Nozl	52.9246	...	1614.5	...	N3-6"
90	Nozl	19.5925	1500	1392.23	...	N4-3"
90	Nozl	17.0902	1385.64	1358.16	800	N5-2"
90	Nozl	17.0893	-1400	1423.22	...	K3-2"
90	Nozl	17.0893	-1503.51	1358.16	547.232	K2-2"
90	Wght	1500	2121.32	1400	-2121.32	PSV1-3"
90	Wght	1500	2598.08	1400	1500	PSV2-3"
80	Trun	545.944	...	1200	...	Trunnions

Total Weight of Each Detail Type:

Platforms	22525.7
Liquid	352.3
Cladding & Overlay	7142.3



Nozzles 746.2
Weights 5375.0
Trunnions 545.9

Sum of the Detail Weights 36687.4 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	81369.9	81369.9	81369.9	81369.9	81369.9	81369.9
Cladding	7142.3	7142.3	7142.3	7142.3	7142.3	7142.3
Nozzles	746.2	746.2	746.2	746.2	746.2	746.2
Wld Weights	2375.0	2375.0	2375.0	2375.0	2375.0	2375.0
Trunnions	545.9	545.9	545.9	545.9	545.9	545.9
Platforms	...	...	...	22525.7	22525.7	22525.7
Empty Weights	...	...	...	3000.0	3000.0	...
Test Weights	...	3000.0	...	...	...	...
Ope Weights	...	...	...	...	...	3000.0
Ope. Liquid	...	...	...	...	...	352.3
Test Liquid	...	353079.3	...	...	...	...
Totals	92179.3	448258.6	92179.3	117705.0	117705.0	118057.3

Field Installation Options:

Platforms installed before lifting.

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	92179.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	448258.6 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	92179.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	117705.0 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	117705.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	118057.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	469892.9 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	49427.7 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	191566.1 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	50336.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	296783	294433
20	30	174453	173322
30	40	344077	337270
40	50	524206	518928
50	60	524206	518928
60	70	524206	518928
70	80	349471	345952
80	90	331997	328655
90	100	344077	337270

Total	3413476.2	3373687.0 cm ²
Total	3674.2	3631.4 ft ²



Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	9270.23	9270.23	9270.23	...	9270.23
20	30	3420.68	3420.68	3420.68	...	3420.68
30	40	8731.44	15948.4	38955.8	...	15925.5
40	50	12133.9	13451.5	84779.9	0.57337	13380.2
50	60	12122.5	16626.6	87955	7247.43	16555.4
60	70	12738.7	17242.8	88571.2	5533.31	17171.6
70	80	8074.07	8952.42	56504.7	...	8904.92
80	90	9708.71	13729.8	58904.4	7336.16	13684.6
90	100	11836.8	19414.9	42422.3	7082.69	19391.9

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	117705	118057	470784
20	30	108435	108787	461514
30	40	105014	105366	458093
40	50	89088.6	89417.9	419138
50	60	75708.4	75966.5	334358
60	70	59153	59339.9	246403
70	80	41981.5	42097	157831
80	90	33076.6	33144.6	101327
90	100	19391.9	19414.9	42422.3

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	27200.2	27200.2	27200.2
20	30	27200.2	27200.2	27200.2
30	40	27200.2	27200.2	27200.2
40	50	27200.2	27200.2	27200.2
50	60	27199.6	27199.6	27199.6
60	70	19952.2	19952.2	19952.2
70	80	14418.8	14418.8	14418.8
80	90	14418.8	14418.8	14418.8
90	100	7082.69	7082.69	7082.69

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N2-3"	47.82	51.00	82	300	GR 1.1	47.010	24.715	47.817	24.715
K1B-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N6-2"	47.82	51.00	82	300	GR 1.1	53.407	18.318	47.817	18.318
K1A-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
M1-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
N1-6"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N3-6"	47.82	51.00	82	300	GR 1.1	30.080	41.645	47.817	41.645
N4-3"	18.20	19.60	82	150	GR 1.1	...	...	...	...
N5-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K3-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K2-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
Min Rating	18.204	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N2-3"	111.125	0.50	129	0	11.430	24.715	Pass
N6-2"	77.851	0.50	50	0	11.430	18.318	Pass
N3-6"	196.088	0.50	454	0	11.430	41.645	Pass

Evaluating Flange on Nozzle [N2-3"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(129) + 4 \cdot \text{abs}(0)111.125 \leq \pi \cdot 111.125^3 \cdot [(47.817 - 11.430) + .500 \cdot 47.817]$$

$$= 0.20667E+04 \leq 0.26507E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-3" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(47.817, 0.500 \cdot 47.817 + 47.817 - 2066.7 / (\pi \cdot 111.125^3))$$

$$= \min(47.817, 24.715)$$

$$= 24.715 \text{ bars}$$

Evaluating Flange on Nozzle [N6-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(50) + 4 \cdot \text{abs}(0)77.851 \leq \pi \cdot 77.851^3 \cdot [(47.817 - 11.430) + .500 \cdot 47.817]$$

$$= 0.80732E+03 \leq 0.91143E+03 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(47.817, 0.500 \cdot 47.817 + 47.817 - 807.3 / (\pi \cdot 77.851^3))$$

$$= \min(47.817, 18.318)$$

$$= 18.318 \text{ bars}$$

Evaluating Flange on Nozzle [N3-6"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(454) + 4 \cdot \text{abs}(0)196.088 \leq \pi \cdot 196.088^3 \cdot [(47.817 - 11.430) + .500 \cdot 47.817]$$

$$= 0.72659E+04 \leq 0.14564E+05 \text{ Check Passes}$$



Maximum Pressure for Flange on Nozzle N3-6" [Pmax]:
= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
= min(47.817, 0.500 • 47.817 + 47.817 - 7265.9/(π • 196.088³))
= min(47.817, 41.645)
= 41.645 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 10.4791 Hz.

The Natural Frequency for the Vessel (Ope...) is 10.4643 Hz.

The Natural Frequency for the Vessel (Filled) is 5.38394 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm.
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm.
Distance Upwind of Crest Lh	0	mm.
Distance from Crest to the Vessel x	0	mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.2 \cdot 0.92) / (1 + 1.7 \cdot 3.4 \cdot 0.2)))$
 $= \min(0.85, 0.88)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	10.464 Hz
Natural Frequency of Vessel (Empty)	10.479 Hz
Natural Frequency of Vessel (Test)	5.384 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.536
Structure Height to Diameter ratio	3.140
Height to top of Structure	17100.500 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274320$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 850 + 300 = 1150$ mm.
 $= 3.77$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (3.77 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0 \cdot 0 \cdot 0)^2$
 $= 1$

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$
 $= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16 \cdot 0.85 \cdot 0.54 \cdot 122$
 $= 889 \text{ lbs. } [403] \text{ Kgf}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt2	1150.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt-1	2500.0	0.000	0.000	0.000	0.849	1.000	78.120
Bottom Dish	3025.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-1	4550.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell2	7550.0	0.000	0.000	0.000	0.943	1.000	78.120
Shell3	10550.0	0.000	0.000	0.000	1.012	1.000	78.120
shell4	13050.0	0.000	0.000	0.000	1.059	1.000	78.120
shell5	15000.0	0.000	0.000	0.000	1.090	1.000	78.120
Top dish	16567.5	0.000	0.000	0.000	1.113	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
D - Average outer insulated diameter of vessel, mm.
Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
Dr - Average outer insulated of top half of vessel, mm.
f - Natural frequency of vibration (Hertz)
f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
L - Total height of structure mm.
Lc - Total length of conical section(s) of vessel, mm.
tb - Uncorroded plate thickness at bottom of vessel, mm.
V30 - Design Wind Speed provided by user, Km/hr
Vc - Critical wind velocity, Km/hr
Vw - Maximum wind speed at top of structure, Km/hr
W - Total corroded weight of structure, Kgf
Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
Z - Maximum amplitude of vibration at top of vessel, mm.
Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:
:: Total Cone Length / Total Length < 0.5
:: $0/15700 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
:: $[18.2/51.5^2] \cdot 10^4 = 68.8$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.40047E-06$ no chance. [Vp]:
 $= W / (L \cdot Dr^2)$
 $= 115196 / (15700 \cdot 5562^2)$
 $= 0.23718E-06$



Compute the damping factor Df which is a measure of instability [Df]:

$$= W \cdot DI / (L \cdot Dr^2)$$

$$= 115196 \cdot 0.03 / (15700 \cdot 5562^2)$$

$$= 0.71155E-08$$

Compute the critical wind velocity [Vc]:

$$= 3.4 \cdot f \cdot Dr$$

$$= 3.4 \cdot 10.5 \cdot 18.2$$

$$= 649.236 \text{ mile/hr } [1044.815 \text{ Km/hr}]$$

Compute the velocity at the top of the tower [Vw]:

$$= V30 \cdot (L / (30 + \text{BaseHeight}))^{0.143}$$

$$= 60.4 \cdot (51.5 / (30 + 0.98))^{0.143}$$

$$= 64.952 \text{ mile/hr } [104.527 \text{ Km/hr}]$$

Compute the maximum gust velocity using the gust response factor Gh [Vg]:

$$= Vw \cdot Gh$$

$$= 105 \cdot 0.85$$

$$= 88.848 \text{ Km/hr}$$

Since Vc is greater than Vg the dynamic deflection Z, does not need to be computed.

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	30000.00	3000.00	78.12	234.36	1.00
PLAT:[1 OF 1]	30000.00	7050.00	78.12	234.36	1.00
PLAT:[1 OF 1]	30000.00	11050.00	78.12	234.36	1.00
PLAT:[1 OF 1]	30000.00	14650.00	78.12	234.36	1.00
SQR PTL	60000.00	17450.00	78.12	468.72	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Skirt beam	0.00	2000.00	78.12	0.00
Davit-1	0.00	10550.00	78.12	0.00
Nozzle Clad	0.00	9050.00	78.12	0.00
Clip wt	0.00	14050.00	78.12	0.00
PSV1-3"	0.00	17350.00	78.12	0.00
PSV2-3"	0.00	17350.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 10.4643 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1150	6668.4	113363	78.12	443.544
20	30	2500	6663.6	66636	78.12	260.72
30	40	3025	6664.8	3332.4	78.12	268.704
40	50	4550	6674.4	200232	78.12	783.429
50	60	7550	6674.4	200232	78.12	1024.15
60	70	10550	6674.4	200232	78.12	1053.44
70	80	13050	6674.4	133488	78.12	522.286
80	90	15000	6674.4	126814	78.12	747.953
90	100	16567.5	6664.8	76641.9	78.12	842.928

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.020
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\
 SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(560.75)$
 $= 0.410$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/10.5)$
 $= 0.096$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.41, 0.096)$ per 12.8.2
 $= 0.096$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 2.5 \cdot 0.16 \cdot 118057 (1 + 2 \cdot 0.02) / (1.25/1.25) \\
 &= 19251.834 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.16 \cdot 1.25 \cdot 118057 = 37023 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 118057$
 $= 6941.768 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 19251.834 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 118057$
 $= 3702.3 \text{ Kgf}$



Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_{xk} / (\text{Sum of } W_i * h_{ik})$ and k is an exponent which is related
to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 10.4643 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	850	9270.23	146.765	152.781	3702.28
20	30	2200	3420.68	140.168	145.913	3411.56
30	40	2725	15948.4	809.465	461.331	3304.29
40	50	4250	13451.5	1064.81	999.887	2804.15
50	60	7250	16626.6	2245.2	1704.07	2382.31
60	70	10250	17242.8	3291.89	2531.67	1860.9
70	80	12750	8952.42	2126.01	1996.01	1320.16
80	90	14700	13729.8	3759.19	2767.19	1039.42
90	100	15675	19414.9	5668.34	3597.52	608.85



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	5947.16	19251.8	54602.7	231765
20	30	2200	5503.62	19105.1	44869.4	199161
30	40	2725	5242.9	18964.9	39496	180125
40	50	4250	4974.19	18155.4	39240.6	179197
50	60	7250	4190.76	17090.6	25492.9	126327
60	70	10250	3166.61	14845.4	14456.6	78422.1
70	80	12750	2113.17	11553.5	6536.77	38822.9
80	90	14700	1590.88	9427.53	2832.65	17841.4
90	100	15675	842.928	5668.34	520.485	3500.04

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	5947.16	0.00065726	0.0025477	0.29025E-05
20	30	5503.62	0.004231	0.0063593	0.46829E-05
30	40	5242.9	0.0064768	0.006595	0.47463E-05
40	50	4974.19	0.014848	0.025079	0.73885E-05
50	60	4190.76	0.036885	0.049901	0.90080E-05
60	70	3166.61	0.063811	0.078348	0.98485E-05
70	80	2113.17	0.088277	0.098331	0.10100E-04
80	90	1590.88	0.10795	0.11762	0.10181E-04
90	100	842.928	0.11787	0.11813	0.10181E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	1252.66	7829.14
20	30	1251.76	7823.51
30	40	1251.99	7824.92
40	50	1253.79	7836.19
50	60	1253.79	7836.19
60	70	1253.79	7836.19
70	80	1253.79	7836.19
80	90	1253.79	7836.19
90	100	1251.99	7824.92

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 78.500 Actual deflection : 0.118 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	3825.52	3652.29	527268576	503573632
20	30	3129.97	2956.64	431709408	407950240
30	40	4169.53	3996.54	573957504	550342016
40	50	4867.97	4694.98	670111936	646530368
50	60	4867.97	4694.98	670111936	646530368
60	70	4867.97	4694.98	670111936	646530368
70	80	4867.97	4694.98	670111936	646530368
80	90	4867.97	4694.98	670111936	646530368
90	100	4169.53	3996.54	573957504	550342016



Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-100.24	-100.24
20	30	82.7985	140.397	-91.9839	-91.9839
30	40	165.597	280.795	-103.631	-103.631
40	50	140.757	238.676	-109.235	-109.235
50	60	140.757	238.676	-109.235	-109.235
60	70	140.757	238.676	-109.235	-109.235
70	80	140.757	238.676	-109.235	-109.235
80	90	140.757	238.676	-109.235	-109.235
90	100	165.597	280.795	-103.631	-103.631

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	65.5628	...	85.2316
40	50	55.8172	...	72.5624
50	60	55.8172	...	72.5624
60	70	55.8172	...	72.5624
70	80	55.8172	...	72.5624
80	90	55.8172	...	72.5624
90	100	65.5628	...	85.2316

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-3.16052	-3.16998	-12.6411	0.5297	0.5297
20	30	-3.59666	-3.60834	-15.3079	0.65386	0.65386
30	40	-2.57687	-2.57687	-2.57687	0.48468	0.48468
40	50	-1.86087	-1.86087	-1.86087	0.41257	0.41257
50	60	-1.58139	-1.58139	-1.58139	0.41257	0.41257
60	70	-1.23558	-1.23558	-1.23558	0.30264	0.30264
70	80	-0.8769	-0.8769	-0.8769	0.21871	0.21871
80	90	-0.6909	-0.6909	-0.6909	0.21871	0.21871
90	100	-0.47584	-0.47584	-0.47584	0.12621	0.12621

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.5297	1.06334	4.5134	0.53167	...
20	30	0.65386	1.07861	4.78758	0.5393	...
30	40	0.48468	0.70379	3.20968	0.35189	...
40	50	0.41257	0.59521	2.71808	0.2976	...
50	60	0.41257	0.38668	1.91614	0.19334	...
60	70	0.30264	0.21928	1.18951	0.10964	...
70	80	0.21871	0.09915	0.58887	0.049575	...
80	90	0.21871	0.042966	0.27062	0.021483	...
90	100	0.12621	0.0092746	0.062368	0.0046373	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	3.28945
20	30	...	...	...	3.47692
30	40	...	...	...	2.32553

40	50	...	...	...	1.96906
50	60	...	...	...	1.3731
60	70	...	...	...	0.83867
70	80	...	...	...	0.40061
80	90	...	...	...	0.17725
90	100	...	...	...	0.039583

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	0.09941
20	30	...	0.11316
30	40	...	0.081082
40	50	...	0.058573
50	60	...	0.049761
60	70	...	0.03887
70	80	...	0.027575
80	90	...	0.021711
90	100	...	0.01494

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
EP = Longitudinal Stress due to External Pressure
HP = Longitudinal Stress due to Hydrotest Pressure
NP = No Pressure
EW = Longitudinal Stress due to Weight (No Liquid)
OW = Longitudinal Stress due to Weight (Operating)
HW = Longitudinal Stress due to Weight (Hydrotest)
WI = Bending Stress due to Wind Moment (Operating)
EQ = Bending Stress due to Earthquake Moment (Operating)
EE = Bending Stress due to Earthquake Moment (Empty)
HI = Bending Stress due to Wind Moment (Hydrotest)
HE = Bending Stress due to Earthquake Moment (Hydrotest)
WE = Bending Stress due to Wind Moment (Empty) (no CA)
WF = Bending Stress due to Wind Moment (Filled) (no CA)
CW = Longitudinal Stress due to Weight (Empty) (no CA)
VO = Bending Stress due to Vortex Shedding Loads (Ope)
VE = Bending Stress due to Vortex Shedding Loads (Emp)
VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
FW = Axial Stress due to Vertical Forces for the Wind Case
FS = Axial Stress due to Vertical Forces for the Seismic Case
BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.75	100.24		0.0474
20		82.80	-5.33	91.98		0.0579
30		165.60	-3.77	103.63		0.0363
40		140.76	-2.87	109.23		0.0263
50		140.76	-2.38	109.23		0.0218
60		140.76	-1.76	109.23		0.0161
70		140.76	-1.19	109.23		0.0109
80		140.76	-0.95	109.23		0.0087
90		165.60	-0.61	103.63		0.0059

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.76	95.76	-6.88	100.24	0.0079	0.0686
20	0.65	82.80	-7.61	91.98	0.0078	0.0828
30	0.31	165.60	-5.31	103.63	0.0019	0.0512
40	0.58	140.76	-4.18	109.23	0.0041	0.0383
50	0.25	140.76	-3.32	109.23	0.0018	0.0304
60		140.76	-2.34	109.23		0.0214
70		140.76	-1.47	109.23		0.0134
80		140.76	-1.07	109.23		0.0098
90		165.60	-0.63	103.63		0.0060



Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.76	100.24		0.0475
20		82.80	-5.34	91.98		0.0581
30		165.60	-3.77	103.63		0.0363
40		140.76	-2.87	109.23		0.0263
50		140.76	-2.38	109.23		0.0218
60		140.76	-1.76	109.23		0.0161
70		140.76	-1.19	109.23		0.0109
80		140.76	-0.95	109.23		0.0087
90		165.60	-0.61	103.63		0.0059

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.97	95.76	-8.11	100.24	0.0206	0.0809
20	1.95	82.80	-8.94	91.98	0.0235	0.0972
30	1.20	165.60	-6.19	103.63	0.0072	0.0597
40	1.33	140.76	-4.93	109.23	0.0094	0.0452
50	0.80	140.76	-3.86	109.23	0.0057	0.0353
60	0.30	140.76	-2.69	109.23	0.0021	0.0246
70		140.76	-1.66	109.23		0.0152
80		140.76	-1.16	109.23		0.0106
90		165.60	-0.65	103.63		0.0063

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-13.70	100.24		0.1367
20		140.40	-16.50	91.98		0.1794
30		280.79	-3.41	103.63		0.0329
40		238.68	-2.57	109.23		0.0235
50		238.68	-2.19	109.23		0.0200
60		238.68	-1.65	109.23		0.0151
70		238.68	-1.15	109.23		0.0105
80		238.68	-0.93	109.23		0.0085
90		280.79	-0.61	103.63		0.0059

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-13.17	100.24		0.1314
20		140.40	-15.96	91.98		0.1735
30		280.79	-3.06	103.63		0.0295
40		238.68	-2.27	109.23		0.0208
50		238.68	-1.99	109.23		0.0183
60		238.68	-1.54	109.23		0.0141
70		238.68	-1.10	109.23		0.0100
80		238.68	-0.91	109.23		0.0083
90		280.79	-0.60	103.63		0.0058

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.76	100.24		0.0475
20		82.80	-5.34	91.98		0.0581
30	64.17	165.60		103.63	0.3875	
40	54.96	140.76		109.23	0.3905	
50	55.04	140.76		109.23	0.3910	
60	55.10	140.76		109.23	0.3915	
70	55.26	140.76		109.23	0.3926	
80	55.39	140.76		109.23	0.3935	



90 65.22 165.60 103.63 0.3939

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.97	95.76	-8.11	100.24	0.0206	0.0809
20	1.95	82.80	-8.94	91.98	0.0235	0.0972
30	66.76	165.60		103.63	0.4032	
40	57.15	140.76		109.23	0.4060	
50	56.61	140.76		109.23	0.4022	
60	56.11	140.76		109.23	0.3986	
70	55.78	140.76		109.23	0.3963	
80	55.64	140.76		109.23	0.3953	
90	65.29	165.60		103.63	0.3943	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.76	100.24		0.0475
20		82.80	-5.34	91.98		0.0581
30		165.60	-3.77	103.63		0.0363
40		140.76	-2.87	109.23		0.0263
50		140.76	-2.38	109.23		0.0218
60		140.76	-1.76	109.23		0.0161
70		140.76	-1.19	109.23		0.0109
80		140.76	-0.95	109.23		0.0087
90		165.60	-0.61	103.63		0.0059

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.97	95.76	-8.11	100.24	0.0206	0.0809
20	1.95	82.80	-8.94	91.98	0.0235	0.0972
30	1.20	165.60	-6.19	103.63	0.0072	0.0597
40	1.33	140.76	-4.93	109.23	0.0094	0.0452
50	0.80	140.76	-3.86	109.23	0.0057	0.0353
60	0.30	140.76	-2.69	109.23	0.0021	0.0246
70		140.76	-1.66	109.23		0.0152
80		140.76	-1.16	109.23		0.0106
90		165.60	-0.65	103.63		0.0063

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-13.70	100.24		0.1367
20		140.40	-16.50	91.98		0.1794
30	83.49	280.79		103.63	0.2973	
40	71.41	238.68		109.23	0.2992	
50	71.59	238.68		109.23	0.2999	
60	71.74	238.68		109.23	0.3006	
70	71.95	238.68		109.23	0.3015	
80	72.11	238.68		109.23	0.3021	
90	84.89	280.79		103.63	0.3023	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-13.17	100.24		0.1314
20		140.40	-15.96	91.98		0.1735
30	83.14	280.79		103.63	0.2961	
40	71.11	238.68		109.23	0.2980	
50	71.39	238.68		109.23	0.2991	
60	71.63	238.68		109.23	0.3001	
70	71.90	238.68		109.23	0.3013	



80	72.09	238.68	109.23	0.3020
90	84.88	280.79	103.63	0.3023

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.69	100.24		0.0368
20		82.80	-4.25	91.98		0.0462
30	63.47	165.60		103.63	0.3833	
40	54.37	140.76		109.23	0.3863	
50	54.65	140.76		109.23	0.3882	
60	54.88	140.76		109.23	0.3899	
70	55.16	140.76		109.23	0.3919	
80	55.35	140.76		109.23	0.3932	
90	65.21	165.60		103.63	0.3938	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.03	100.24		0.0302
20		82.80	-3.41	91.98		0.0371
30	63.09	165.60		103.63	0.3810	
40	54.02	140.76		109.23	0.3838	
50	54.29	140.76		109.23	0.3857	
60	54.63	140.76		109.23	0.3881	
70	54.97	140.76		109.23	0.3905	
80	55.15	140.76		109.23	0.3918	
90	65.11	165.60		103.63	0.3932	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.70	100.24		0.0369
20		82.80	-4.26	91.98		0.0463
30	63.47	165.60		103.63	0.3833	
40	54.37	140.76		109.23	0.3863	
50	54.65	140.76		109.23	0.3882	
60	54.88	140.76		109.23	0.3899	
70	55.16	140.76		109.23	0.3919	
80	55.35	140.76		109.23	0.3932	
90	65.21	165.60		103.63	0.3938	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.69	100.24		0.0368
20		82.80	-4.25	91.98		0.0462
30	63.47	165.60		103.63	0.3833	
40	54.37	140.76		109.23	0.3863	
50	54.65	140.76		109.23	0.3882	
60	54.88	140.76		109.23	0.3899	
70	55.16	140.76		109.23	0.3919	
80	55.35	140.76		109.23	0.3932	
90	65.21	165.60		103.63	0.3938	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.70	100.24		0.0369
20		82.80	-4.26	91.98		0.0463
30		165.60	-3.06	103.63		0.0295
40		140.76	-2.27	109.23		0.0208
50		140.76	-1.99	109.23		0.0183
60		140.76	-1.54	109.23		0.0141



70	140.76	-1.10	109.23	0.0100
80	140.76	-0.91	109.23	0.0083
90	165.60	-0.60	103.63	0.0058

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.60	83.53		0.0431
20		69.00	-4.15	76.65		0.0541
30	63.55	138.00		86.36	0.4605	
40	54.43	117.30		91.03	0.4640	
50	54.70	117.30		91.03	0.4663	
60	54.92	117.30		91.03	0.4682	
70	55.19	117.30		91.03	0.4705	
80	55.37	117.30		91.03	0.4720	
90	65.23	138.00		86.36	0.4727	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.60	83.53		0.0431
20		69.00	-4.15	76.65		0.0541
30		138.00	-2.98	86.36		0.0345
40		117.30	-2.21	91.03		0.0243
50		117.30	-1.94	91.03		0.0214
60		117.30	-1.50	91.03		0.0165
70		117.30	-1.07	91.03		0.0117
80		117.30	-0.89	91.03		0.0098
90		138.00	-0.59	86.36		0.0068

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.70	100.24		0.0369
20		82.80	-4.26	91.98		0.0463
30	63.47	165.60		103.63	0.3833	
40	54.37	140.76		109.23	0.3863	
50	54.65	140.76		109.23	0.3882	
60	54.88	140.76		109.23	0.3899	
70	55.16	140.76		109.23	0.3919	
80	55.35	140.76		109.23	0.3932	
90	65.21	165.60		103.63	0.3938	

Absolute Maximum of the all of the Stress Ratio's 0.4727

Governing Element: Top dish

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt2	...	-3.1	0.5	...	...	+ Tens
Skirt2	...	-3.1	-0.5	...	-3.6	- Comp
Skirt-1	...	-3.5	0.7	...	...	+ Tens
Skirt-1	...	-3.5	-0.7	...	-4.1	- Comp
Bottom Dish	65.6	-2.5	0.5	...	63.6	+ Tens
Bottom Dish	65.6	-2.5	-0.5	...	...	- Comp
Shell-1	55.8	-1.8	0.4	...	54.4	+ Tens
Shell-1	55.8	-1.8	-0.4	...	...	- Comp
Shell12	55.8	-1.5	0.4	...	54.7	+ Tens
Shell12	55.8	-1.5	-0.4	...	...	- Comp



Shell3	55.8	-1.2	0.3	...	54.9	+ Tens
Shell3	55.8	-1.2	-0.3	...	...	- Comp
shell14	55.8	-0.8	0.2	...	55.2	+ Tens
shell14	55.8	-0.8	-0.2	...	...	- Comp
shell15	55.8	-0.7	0.2	...	55.4	+ Tens
shell15	55.8	-0.7	-0.2	...	...	- Comp
Top dish	65.6	-0.5	0.1	...	65.2	+ Tens
Top dish	65.6	-0.5	-0.1	...	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

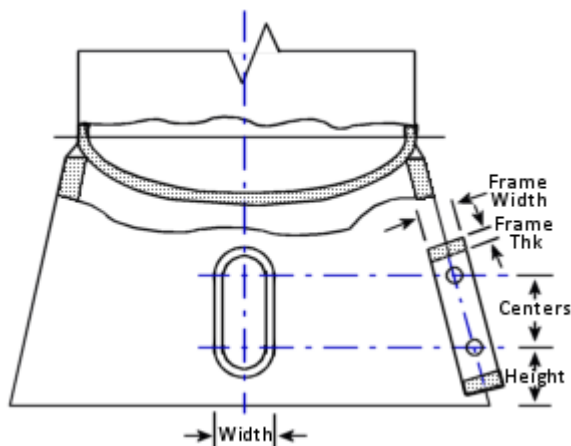
Total vessel (empty) weight	Fg	106615.367 Kg
Head Weight below section 2-2	DeltaFg	15925.463 Kg
Weight of vessel contents (liquid)	Ff	352.263 Kg
Vessel Gross Weight	Fg + Ff	106967.633 Kg
Skirt Mean Diameter	Dz	5536.000 mm.
Skirt Analysis Thickness (Corroded)	ez	17.000 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kg
Moment at section 1-1	M1	180125.266 Kg-m.
Applied Moment - Bottom of Skirt	Mb	199160.641 Kg-m.
Weight at Bottom of Skirt	W	108787.023 Kg

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	5531.000 mm.
Head Corroded Thickness	eb	23.000 mm.

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 1378 \text{ mm.} \\
 K &= D_i / (2 * h_i) = 5508 / (2 * 1378) = 2 \\
 R &= D_i (0.44 * K + 0.02) = 5508 * (0.44 * 2 + 0.02) = 4955 \text{ mm.} \\
 r &= D_i ((0.5 / K) - 0.08) = 5508 * ((0.5 / 2) - 0.08) = 937 \text{ mm.}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	113265240.000 cm ⁴
Moment of Inertia	Iyy =	113265240.000 cm ⁴
Section Modulus	W =	409195648.000 mm. ³
Section Metal Area	A =	2956.616 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(113265240, 113265240) / 2957)} \\
 &= 1957 \text{ mm.}
 \end{aligned}$$



Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= r_g / L \\ &= 1957/1000 \\ &= 1.957 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 180125 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

$$\begin{aligned} &\text{Downwind Side:} \\ &= -M / Z - W / A \\ &= -180125/1636782592 - 108787/2957 \\ &= -4.69 \text{ N./mm}^2 \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +180125/1636782592 - 108787/2957 \\ &= -2.53 \text{ N./mm}^2 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\ &= 0 - 106615 - 352 + 4 \cdot 180125/5536 \\ &= 23178 \text{ Kgf} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\ &= 0 - 106615 - 352 - 4 \cdot 180125/5536 \\ &= -237113 \text{ Kgf} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (23178 + 15925 + 352) / (\pi \cdot 5531 \cdot 23) + 11 \cdot 5531 / (4 \cdot 23) \\ &= 67.03 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-237113 + 15925 + 352) / (\pi \cdot 5531 \cdot 23) + 11 \cdot 5531 / (4 \cdot 23) \\ &= 60.64 \text{ N./mm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (352 + 15925) / (\pi \cdot 5531 \cdot 23) + 11 \cdot 5531 / (4 \cdot 23) \\ &= 66.46 \text{ N./mm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= F_{zp} / (\pi \cdot D_z \cdot e_z) \\ &= 23178 / (\pi \cdot 5536 \cdot 17) \\ &= 0.77 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= F_{zq} / (\pi \cdot D_z \cdot e_z) \\ &= -237113 / (\pi \cdot 5536 \cdot 17) \\ &= -7.86 \text{ N./mm}^2 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(5531 + 23 - 5536 + 17) / (937 + 23)$$

$$= 0.98$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.98) = 11 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(23^2 + 17^2 + 2 \cdot 23 \cdot 17 \cdot 0.98)}$$

$$= 19.9 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 19.91 \cdot 23178.1 = 461.50 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 19.91 \cdot -237113.4 = -4721.19 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(23/17)^2$$

$$= 0.526$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.53 \cdot 6 \cdot 462 / (\pi \cdot 5531 \cdot 23^2)$$

$$= 1.55 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.53 \cdot 6 \cdot -4721 / (\pi \cdot 5531 \cdot 23^2)$$

$$= -15.89 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.53 \cdot 6 \cdot 462 / (\pi \cdot 5536 \cdot 17^2)$$

$$= 2.84 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.53 \cdot 6 \cdot -4721 / (\pi \cdot 5536 \cdot 17^2)$$

$$= -29.05 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.91/218 = 0.0042$$

$$y = 125 \cdot eb/D_b = 125 \cdot 23/5531 = 0.52$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.52 + 3.994 = 4.16$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \cdot ((\gamma / \gamma_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (11 + 0) \cdot 5531 \cdot ((11/40) \cdot 4.16 - 1) / (4 \cdot 23)$$

$$= 9.26 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 67 - 1.55 + 9.26$$

$$= 74.73 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 67 + 1.55 - 9.26$$

$$= 59.32 \text{ N./mm}^2$$



Stress [σ_{1qitot}]:
 $= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$
 $= 60.6 - -15.9 + 9.26$
 $= 85.78 \text{ N./mm}^2$

Stress [σ_{1qotot}]:
 $= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$
 $= 60.6 + -15.9 - 9.26$
 $= 35.49 \text{ N./mm}^2$

Stress [σ_{2pitot}]:
 $= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$
 $= 66.5 + 1.55 + 9.26$
 $= 77.27 \text{ N./mm}^2$

Stress [σ_{2potot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 66.5 - 1.55 - 9.26$
 $= 55.64 \text{ N./mm}^2$

Stress [σ_{2qitot}]:
 $= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$
 $= 66.5 + -15.9 + 9.26$
 $= 59.83 \text{ N./mm}^2$

Stress [σ_{2qotot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 66.5 - -15.9 - 9.26$
 $= 73.08 \text{ N./mm}^2$

Stress [σ_{3pitot}]:
 $= \sigma_{3pm} - \sigma_{3pb}$
 $= 0.77 - 2.84$
 $= -2.07 \text{ N./mm}^2$

Stress [σ_{3potot}]:
 $= \sigma_{3pm} + \sigma_{3pb}$
 $= 0.77 + 2.84$
 $= 3.61 \text{ N./mm}^2$

Stress [σ_{3qitot}]:
 $= \sigma_{3qm} - \sigma_{3qb}$
 $= -7.86 - -29.1$
 $= 21.19 \text{ N./mm}^2$

Stress [σ_{3qotot}]:
 $= \sigma_{3qm} + \sigma_{3qb}$
 $= -7.86 + -29.1$
 $= -36.92 \text{ N./mm}^2$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5) \cdot (\sigma_{1pm} / f)^2)$
 $= 138(3 - (1/1.5) \cdot (67/138)^2)$
 $= 392.29 \text{ N./mm}^2$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5) \cdot (\sigma_{1qm} / f)^2)$
 $= 138(3 - (1/1.5) \cdot (60.6/138)^2)$
 $= 396.23 \text{ N./mm}^2$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5) \cdot (\sigma_{2pm} / f)^2)$
 $= 138(3 - (1/1.5) \cdot (66.5/138)^2)$
 $= 392.66 \text{ N./mm}^2$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:



$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2) \\ &= 138 (3 - (1/1.5) \cdot (66.5/138)^2) \\ &= 392.66 \text{ N./mm}^2 \end{aligned}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2) \\ &= 138 (3 - (1/1.5) \cdot (0.77/138)^2) \\ &= 413.99 \text{ N./mm}^2 \end{aligned}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2) \\ &= 138 (3 - (1/1.5) \cdot (-7.86/138)^2) \\ &= 413.69 \text{ N./mm}^2 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	74.73	392.29	Pass	46
Stot1po	59.32	392.29	Pass	47
Stot1qi	85.78	396.23	Pass	48
Stot1qo	35.49	396.23	Pass	49
Stot2pi	77.27	392.66	Pass	50
Stot2po	55.64	392.66	Pass	51
Stot2qi	59.83	392.66	Pass	52
Stot2qo	73.08	392.66	Pass	53
Stot3pi	-2.07	413.99	Pass	54
Stot3po	3.61	413.99	Pass	55
Stot3qi	21.19	413.69	Pass	56
Stot3qo	-36.92	413.69	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
 Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	10334.345 mm.
Center of Gravity of the Liquid	9199.999 mm.
Center of Gravity of the Lining	9200.000 mm.
Center of Gravity of the Nozzles	11191.629 mm.
Center of Gravity of the Added Weights (Operating)	13663.256 mm.
Center of Gravity of the Added Weights (Empty)	13663.256 mm.
Center of Gravity of the Trunnions	14950.000 mm.
Center of Gravity of Bare Shell New and Cold	8025.406 mm.
Center of Gravity of Bare Shell Corroded	8021.396 mm.
Vessel CG in the Operating Condition	8866.974 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	8439.397 mm.
Vessel CG in the Test Condition	9048.201 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	15700.00 mm.
Total vessel weight (No Liquid) Twt	117704.99 Kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	176557.48 Kgf
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	14950.00 mm.
Design Reaction force at the tailing lug	77668.61 Kgf
Design Reaction force at the Trunnions	98888.88 Kgf
CG distance during field lifting operation	8439.40 mm.
CG distance from tailing lug	8289.40 mm.
CG distance from the nearer lifting lug/trunnion	6510.60 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	3.23	6950.00	92.06 (UG-23)
Shear	1.58	14890.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	35749.6	0.9
2300.00	127799.2	2.9	28805.3	0.9
3350.00	160106.2	2.3	18859.5	0.4
8150.00	217981.6	3.2	20057.3	0.4
12550.00	104149.2	1.5	56943.8	1.1
15650.00	13410.4	0.2	3165.2	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.04 at elevation 2500 mm. - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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Trunnion Analysis:

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm ²
Trunnion Outside Diameter	Tod 550.000 mm.
Trunnion Thickness	Tthk 22.0000 mm.
Trunnion Length	Tl 530.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 900.000 mm.
Pad Thickness	Thkp 28.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 24.0000 mm.
Bail/Sling Width	Tt 32.0000 mm.
Sling Angle from Horizontal	0.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2

Trunnion Force and Stress Results for different Lift Angles (Kgf&N./mm²):

Bending Stress Allowable:	172.9 N./mm ²
Shear Stress Allowable:	104.8 N./mm ²
Bearing Stress Allowable:	196.5 N./mm ²

Angle	Trunnion Lifting Load	Bending Stress	Shear Stress	Bearing Stress	Unity Check	Pass Fail
0	49444.4	53.8	13.3	27.6	0.438	Pass
5	50129.2	54.5	13.5	27.9	0.444	Pass
10	50800.3	55.3	13.7	28.3	0.450	Pass
15	51468.1	56.0	13.8	28.7	0.456	Pass
20	52142.9	56.7	14.0	29.1	0.462	Pass
25	52835.4	57.5	14.2	29.4	0.468	Pass
30	53557.3	58.3	14.4	29.8	0.474	Pass
35	54322.6	59.1	14.6	30.3	0.481	Pass
40	55148.1	60.0	14.8	30.7	0.488	Pass
45	56055.6	61.0	15.1	31.2	0.496	Pass
50	57074.3	62.1	15.3	31.8	0.506	Pass
55	58244.7	63.4	15.7	32.5	0.516	Pass
60	59626.4	64.9	16.0	33.2	0.528	Pass
65	61310.2	66.7	16.5	34.2	0.543	Pass
70	63443.8	69.0	17.0	35.4	0.562	Pass
75	66285.0	72.1	17.8	36.9	0.587	Pass
80	70329.6	76.5	18.9	39.2	0.623	Pass
85	76669.4	83.4	20.6	42.7	0.679	Pass
90	88278.7	96.1	23.7	49.2	0.782	Pass

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 58852 • 1.5
 = 88278.7 Kgf

Trunnion Section Modulus [Z]:
 = $\pi (Tod^4 - Tid^4) / (32 \cdot Tod)$
 = $\pi (550^4 - 506^4) / (32 \cdot 550)$
 = 4632386.5 mm.³

Trunnion Bending Moment [Tmom]:
 = $(Tl - 0.5 \cdot Tt) / 9.807 \cdot Tlf$
 = $(530 - 0.5 \cdot 32) / 9.807 \cdot 88279$
 = 45376.2 Kg-m.



Trunnion Calculated Bending Stress [Sbc]:

$$\begin{aligned} &= T_{mom} \cdot 9.807/Z \\ &= 45376 \cdot 9.807/4632386 \\ &= 96.06 \text{ N./mm}^2 \text{ [Passed] } Sbc < \min(0.66 \cdot SY \cdot Occfac, 0.75 \cdot Sy) (172.92) \end{aligned}$$

Trunnion Cross Section Area [Txsa]:

$$\begin{aligned} &= \pi/4(T_{od}^2 - T_{id}^2) \\ &= \pi/4(550^2 - 506^2) \\ &= 364.9 \text{ cm}^2 \end{aligned}$$

Trunnion Calculated Shear Stress [Ssc]:

$$\begin{aligned} &= T_{lf}/T_{xsa} \\ &= 88279/365 \\ &= 23.72 \text{ N./mm}^2 \text{ [Passed] } Ssc < 0.4 \cdot Sy \cdot Occfac (104.80) \end{aligned}$$

Trunnion Unity Check [Uchk]:

$$\begin{aligned} &= Sbc/Sba + Ssc/Ssa \\ &= 96.1/173 + 23.7/105 \\ &= 0.782 \text{ [Passed] } \end{aligned}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$\begin{aligned} &= \pi(T_{od}/2)^2 \\ &= \pi(550/2)^2 \\ &= 237582.7 \text{ mm}^2 \end{aligned}$$

Shear Stress in Weld due to Bending Moment [fs]:

$$\begin{aligned} &= T_{mom}/S_w \\ &= 45376/237583 \\ &= 190.987 \text{ Kgf/mm.} \end{aligned}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$\begin{aligned} &= \max(10\text{mm}, f_s/(0.4 \cdot Sy)) \\ &= \max(10\text{mm}, 191/(0.4 \cdot 262)) \\ &= 17.872 \text{ mm.} \end{aligned}$$

Trunnion Bearing Contact Area [Tbca]:

$$\begin{aligned} &= T_{od} \cdot T_t \\ &= 550 \cdot 32 \\ &= 176.0 \text{ cm}^2 \end{aligned}$$

Trunnion Bearing Stress [Sbec]:

$$\begin{aligned} &= T_{lf}/T_{bca} \\ &= 88279/176 \\ &= 49.19 \text{ N./mm}^2 \text{ [Passed] } Sbec < \min(0.75 \cdot SY \cdot Occfac, 0.9 \cdot Sy) (196.50) \end{aligned}$$

Summary of Results:

Trunnion Unity Check Value: 0.7819 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	96.06	172.92	Pass
Shear Stress	23.72	104.80	Pass
Bearing Stress	49.19	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).



Loads on shell and WRC 107 results at different Lift Angles (Kgf & Kg-m.):

Angle	PT	PL	MC	ML	Rat1	Rat2	P/F
0	49444.4	0.0	25414.9	0.0	0.488	0.829	Pass
5	49938.5	4369.0	25668.9	2245.7	0.493	0.837	Pass
9	50042.9	7926.0	25722.5	4074.1	0.494	0.839	Pass
10	50028.5	8821.4	25715.2	4534.3	0.494	0.838	Pass
15	49714.4	13320.9	25553.7	6847.1	0.491	0.833	Pass
20	48998.3	17833.9	25185.6	9166.8	0.484	0.821	Pass
25	47885.1	22329.2	24613.4	11477.4	0.473	0.803	Pass
30	46382.0	26778.7	23840.8	13764.5	0.459	0.778	Pass
35	44498.5	31158.2	22872.7	16015.6	0.441	0.747	Pass
40	42245.9	35448.5	21714.8	18220.9	0.419	0.709	Pass
45	39637.3	39637.3	20374.0	20374.0	0.394	0.666	Pass
50	36686.6	43721.4	18857.3	22473.3	0.365	0.617	Pass
55	33407.8	47711.3	17172.0	24524.1	0.373	0.563	Pass
60	29813.2	51638.0	15324.3	26542.4	0.403	0.504	Pass
65	25910.8	55565.9	13318.4	28561.5	0.432	0.445	Pass
70	21699.1	59617.7	11153.5	30644.1	0.463	0.475	Pass
75	17155.8	64026.4	8818.3	32910.2	0.497	0.509	Pass
80	12212.6	69261.1	6277.4	35600.9	0.537	0.550	Pass
85	6682.2	76377.7	3434.7	39258.9	0.592	0.605	Pass
90	0.0	88278.7	0.0	45376.2	0.684	0.699	Pass

- PT - Transverse load per lug
- PL - Longitudinal load per lug
- MC - Circumferential moment in shell
- ML - Longitudinal moment in shell
- Rat1 - Computed WRC 107 stress ratio at trunnion edge
- Rat2 - Computed WRC 107 stress ratio at pad edge

Worst case combination loads were at: 9.0 degrees from horizontal.

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	0.0000	mm.
Vessel Diameter	Dv	5506.000	mm.
Vessel Thickness	Tv	28.000	mm.
Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	550.000	mm.
Nozzle Thickness	Tn	22.000	mm.
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	28.000	mm.
Diameter of Reinforcing Pad	Dpad	948.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

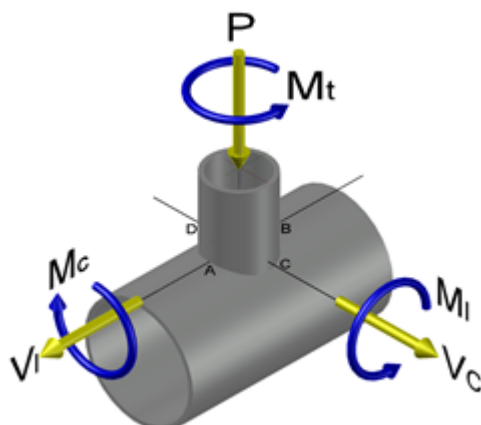
Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	7926.0	Kgf
Circumferential Shear (SUS)	Vc	50042.9	Kgf
Circumferential Moment (SUS)	Mc	25722.5	Kg-m.
Longitudinal Moment (SUS)	Ml	4074.1	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.



WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Worst case combination loads were at: 9.0 degrees from horizontal.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 550.0 + 2 \cdot 1.65 \cdot \sqrt{(2767.0 (28.0 - 0))}$
 $= 1468.539 \text{ mm.}$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	50042.9	Kgf
Longitudinal Shear	VL	7926.0	Kgf
Circumferential Moment	MC	25722.5	Kg-m.
Longitudinal Moment	ML	4074.1	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 49.66

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.087	4C	8.749	(A,B)
N(PHI) / (P/Rm)	0.087	3C	7.836	(C,D)
M(PHI) / (P)	0.087	2C1	0.090	(A,B)
M(PHI) / (P)	0.087	1C	0.124	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.087	3A	1.466	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.087	1A	0.097	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.087	3B	5.167	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.087	1B	0.047	(A,B,C,D)
N(x) / (P/Rm)	0.087	3C	7.836	(A,B)
N(x) / (P/Rm)	0.087	4C	8.749	(C,D)
M(x) / (P)	0.087	1C1	0.129	(A,B)
M(x) / (P)	0.087	2C	0.089	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.087	4A	1.995	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.087	2A	0.053	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.087	4B	1.385	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.087	2B	0.075	(A,B,C,D)



Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-9.9	-9.9	9.9	9.9
Circ. Bend. MC		0.0	0.0	0.0	0.0	-194.7	194.7	194.7	-194.7
Circ. Memb. ML		-5.5	-5.5	5.5	5.5	0.0	0.0	0.0	0.0
Circ. Bend. ML		-15.0	15.0	15.0	-15.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-20.5	9.5	20.5	-9.5	-204.6	184.8	204.6	-184.8
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-13.4	-13.4	13.4	13.4
Long. Bend. MC		0.0	0.0	0.0	0.0	-106.8	106.8	106.8	-106.8
Long. Memb. ML		-1.5	-1.5	1.5	1.5	0.0	0.0	0.0	0.0
Long. Bend. ML		-23.9	23.9	23.9	-23.9	0.0	0.0	0.0	0.0
Tot. Long. Str.		-25.4	22.5	25.4	-22.5	-120.2	93.4	120.2	-93.4
Shear VC		10.1	10.1	-10.1	-10.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		10.1	10.1	-10.1	-10.1	-1.6	-1.6	1.6	1.6
Str. Int.		33.4	28.0	33.4	28.0	204.6	184.9	204.6	184.9

Dimensionless Parameters used : $\Gamma = 98.82$

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
$N(\Phi) / (P/R_m)$	0.150	4C	12.956	(A,B)
$N(\Phi) / (P/R_m)$	0.150	3C	7.660	(C,D)
$M(\Phi) / (P)$	0.150	2C1	0.029	(A,B)
$M(\Phi) / (P)$	0.150	1C !	0.066	(C,D)
$N(\Phi) / (MC/(R_m^{**2} * \beta))$	0.150	3A	3.897	(A,B,C,D)
$M(\Phi) / (MC/(R_m^{**2} * \beta))$	0.150	1A	0.068	(A,B,C,D)
$N(\Phi) / (ML/(R_m^{**2} * \beta))$	0.150	3B	9.269	(A,B,C,D)
$M(\Phi) / (ML/(R_m^{**2} * \beta))$	0.150	1B	0.020	(A,B,C,D)
$N(x) / (P/R_m)$	0.150	3C	7.660	(A,B)
$N(x) / (P/R_m)$	0.150	4C	12.956	(C,D)
$M(x) / (P)$	0.150	1C1	0.060	(A,B)
$M(x) / (P)$	0.150	2C !	0.036	(C,D)
$N(x) / (MC/(R_m^{**2} * \beta))$	0.150	4A	8.158	(A,B,C,D)
$M(x) / (MC/(R_m^{**2} * \beta))$	0.150	2A	0.032	(A,B,C,D)
$N(x) / (ML/(R_m^{**2} * \beta))$	0.150	4B	3.820	(A,B,C,D)
$M(x) / (ML/(R_m^{**2} * \beta))$	0.150	2B	0.025	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

	Stress Intensity Values at
--	----------------------------



Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-30.6	-30.6	30.6	30.6
Circ. Bend. MC		0.0	0.0	0.0	0.0	-316.6	316.6	316.6	-316.6
Circ. Memb. ML		-11.5	-11.5	11.5	11.5	0.0	0.0	0.0	0.0
Circ. Bend. ML		-14.5	14.5	14.5	-14.5	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-26.0	2.9	26.0	-2.9	-347.2	286.0	347.2	-286.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-64.0	-64.0	64.0	64.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-146.7	146.7	146.7	-146.7
Long. Memb. ML		-4.7	-4.7	4.7	4.7	0.0	0.0	0.0	0.0
Long. Bend. ML		-18.5	18.5	18.5	-18.5	0.0	0.0	0.0	0.0
Tot. Long. Str.		-23.3	13.8	23.3	-13.8	-210.8	82.7	210.8	-82.7
Shear VC		11.8	11.8	-11.8	-11.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.9	-1.9	1.9	1.9
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		11.8	11.8	-11.8	-11.8	-1.9	-1.9	1.9	1.9
Str. Int.		36.5	25.9	36.5	25.9	347.2	286.0	347.2	286.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-5.5	-5.5	5.5	5.5	-9.9	-9.9	9.9	9.9
Circ. Q (SUS)		-15.0	15.0	15.0	-15.0	-194.7	194.7	194.7	-194.7
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-1.5	-1.5	1.5	1.5	-13.4	-13.4	13.4	13.4
Long. Q (SUS)		-23.9	23.9	23.9	-23.9	-106.8	106.8	106.8	-106.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		10.1	10.1	-10.1	-10.1	-1.6	-1.6	1.6	1.6
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		20.7	20.7	20.7	20.7	14.0	14.0	14.0	14.0
Pm+Pl+Q (Total)		33.4	28.0	33.4	28.0	204.6	184.9	204.6	184.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	20.68	207.00	Passed
Pm+Pl+Q (TOTAL)	204.61	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.



The P_m+P_l+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-11.5	-11.5	11.5	11.5	-30.6	-30.6	30.6	30.6
Circ. Q (SUS)		-14.5	14.5	14.5	-14.5	-316.6	316.6	316.6	-316.6
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-4.7	-4.7	4.7	4.7	-64.0	-64.0	64.0	64.0
Long. Q (SUS)		-18.5	18.5	18.5	-18.5	-146.7	146.7	146.7	-146.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		11.8	11.8	-11.8	-11.8	-1.9	-1.9	1.9	1.9
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		24.5	24.5	24.5	24.5	64.1	64.1	64.1	64.1
Pm+Pl+Q (Total)		36.5	25.9	36.5	25.9	347.2	286.0	347.2	286.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	64.15	207.00	Passed
Pm+Pl+Q (TOTAL)	347.22	413.99	Passed

The P_m+P_l+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The P_m+P_l+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	5557.0000	mm.
Skirt Thickness	STHK	22.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material	SA-36		

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm ²
Basing Yield Stress		262.00	N./mm ²
Inside Diameter of Basing	DI, D4	5389.0000	mm.
Outside Diameter of Basing	DOU, D5	5989.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Root Area of a Single Bolt	Area	21.5100	cm ²
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm ²
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	5773.0000	mm.
Bolt Allowable Shear Stress		82.800	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	22.0000	mm.
Width of Gussets at Top Plate	TWDT	206.0000	mm.
Width of Gussets at Base Plate	BWDT	206.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm ²
Height of Gussets	HG, h1	312.0000	mm.
Distance between Gussets	RG, b6	120.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	60.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	50.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	216.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	90.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	117705.0	Kgf
Operating Weight of Vessel	ROW	118057.3	Kgf
Earthquake Moment on Basing	EQMOM	238653.9	Kg-m.
Wind Moment on Basing	WIMOM	61492.0	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 238654/5773) - 114355) / 24$$



= 2125.0 Kgf

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 61492/5773) - 117705) / 24$$

$$= -3129.1 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/5773) - 118057) / 24$$

$$= -4919.1 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in one Bolt	Abss :	21.5100	cm ²
Area Available in all the Bolts	Abss * RN :	516.2400	cm ²



*There seems to be no tensile load on bolts per the Brownell and Young Method.
 It is possible that this method is encountering a problem. It is strongly
 recommended that you try the simplified analysis method for the basing.*

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 0.38[(2 \cdot 1 \cdot 5773 + 300) / (2 \cdot 1 \cdot 5773)]$$

$$= 0.39 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 216, b = 378, l/b = 0.57166755

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(231, 519) = 519 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 519/173)} + 1$$

$$= 14.3 \text{ mm.}$$

Nomenclature:

a = (Dc-Ds)/2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 216, b = 120, b/l = 0.55555558

As b/l (0.56) is less than 1, inverting b/l = 1.80.

Moment Term, based on geometry [Mo]:

$$= P/(4 \cdot \pi) [1.3(\ln((2\sin(\pi \cdot a/l)/(\pi \cdot g))) + 1) - [(0.7 - g2)P/(4\pi)]$$

$$= 30268/(4 \cdot \pi) [1.3(\ln((2 \cdot 216 \cdot \sin(\pi \cdot 108/216)) /$$

$$(\pi \cdot 54))) + 1] - [(0.7 - 0.04) \cdot 30268/(4 \cdot \pi)]$$

$$= 3746.5251 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot Abs(3747) / 175)} + 1$$

$$= 36.5266 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot lavg) + Ca$$

$$= 30268/(124 \cdot 206) + 1$$

$$= 12.6 \text{ (not less than } 9.52 + 1) \text{ mm.}$$

Bolt spacing [m]:



$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 5773/24 = 756 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$$

$$= 1.76(30268 \cdot 108 / (756 \cdot 350 \cdot 171))^{2/3} \cdot 2778^{1/3} + Ca$$

$$= 19.7 + 1 = 20.7 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 19252 / (21.5 \cdot 24)$$

$$= 3.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	14.2867	mm.
Actual Basing Thickness as entered by user	38.0000	mm.
Required Thickness of Chair Cap	36.5266	mm.
Actual Top Ring Thickness as entered by user	50.0000	mm.
Required Gusset thickness, + CA	12.6104	mm.
Actual Gusset Thickness as entered by user	22.0000	mm.
Required Thickness of Skirt for Local Stress	20.6704	mm.
Given Thickness of Skirt	22.0000	mm.
Required Gusset Height to meet local stress	279.2927	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$

$$= 13.7 \cdot (22 - 1)$$

$$= 29.3 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 29.3 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 2.21 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (Hg + Tta))$$

$$= 30268 / (184 + 2 \cdot (312 + 50))$$

$$= 33.3 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$$

$$= 30268 \cdot 108 / (184 \cdot (362) + 0.6667 \cdot (362)^2)$$

$$= 21.2 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{ Wv^2 + Wh^2 }$$

$$= \sqrt{ 33.3^2 + 21.2^2 }$$

$$= 39.5 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 39.5 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 2.98 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 30268 / (2 \cdot 206)$$

$$= 73.5 \text{ Kg/mm.}$$



Weld Load [Wh]:
 = Bolt Load • e / (2 • Hgt • TopWth)
 = 30268 • 108 / (2 • 362 • 206)
 = 21.9 Kg/mm.

Resultant Weld Load [Wr]:
 = $\sqrt{(Wv^2 + Wh^2)}$
 = $\sqrt{(73.5^2 + 21.9^2)}$
 = 76.7 Kg/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
 = Wr / [(0.4 • Yield) • 2 • 0.707]
 = 76.7 / [(0.4 • 230) • 2 • 0.707]
 = 5.78 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basinger to Skirt Double Fillet Weld Size	6.3500 mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500 mm.
Required Top Plate to Skirt Weld Size	6.3500 mm.
Required Gusset to Top Plate Double Fillet Weld Size	5.7776 mm.

Tailing Lug Stress Calculation Results :

Tail Lug Type:	Dual		
Tail lug yield stress	Syt	262.00	N./mm ²
Tail lug allowable stress	Sat	138.00	N./mm ²
Tail lug thickness	TTL	40.0000	mm.
Pin hole diameter	DPH	60.0000	mm.
Fillet weld size	WS	30.0000	mm.
Tail lug load	TLOAD	38834	Kgf

Decay Length Along the Skirt (e):
 = $0.5 \cdot \sqrt{(DS/2 \cdot Ts)}$
 = $0.5 \cdot \sqrt{(5557/2 \cdot 22)}$
 = 123.62 mm.

Skirt Effective Length :
 = HG + TTA + e
 = 312 + 50 + 124
 = 485.62 mm.

Section	CG (Y) mm.	Width (H) mm.	Area (A) cm ²	Moment Area mm. ³
Top plate, t	192.00	216.00	108.00	2073598.62
Skirt, s	73.00	485.62	106.84	779903.69
Bottom plate, b	150.00	300.00	114.00	1709998.88
TOTAL			328.84	4563501.50

Calculated Overall CG (Y):
 = Mtotal / Atotal
 = 4563501/329
 = 138.78 mm.

Calculated Moment of Inertia (I):
 = (At • Yt² + Topwth³ • TTA/12 + As • Ys² +
 Ts³ • Hs/12 + Ab • Yb² + Hb³ • Tb/12) -
 Mtotal²/Atotal
 = (108 • 192² + 216³ • 50/12 + 107 • 73² +
 22³ • 486/12 + 114 • 150² + 300³ • 38/12) -
 4563501²/329
 = 20617.39 cm⁴



$$\begin{aligned}\alpha &= \arcsin(OS / (DS / 2)) \\ &= \arcsin(1000/(5557/2)) \\ &= 0.368 \text{ rad. (21.094 deg.)}\end{aligned}$$

$$\begin{aligned}K &= \sin(\alpha) \cdot (\sin(\alpha) - \pi + \alpha) + k2 \cdot (1 + \cos(\alpha)) \\ &= \sin(0.37) \cdot (\sin(0.37) - 3.14 + 0.37) + 1 \cdot (1 + \cos(0.37)) \\ &= 1.06\end{aligned}$$

Required Section Modulus (Zr) - per Roark's Table 17 ref 4

$$\begin{aligned}&= K \cdot TLOAD \cdot (DI/2 + Y) / (0.66 \cdot Sy \cdot \pi) \\ &= 1.06 \cdot 38834 \cdot (5389/2 + 139) / (0.66 \cdot 262 \cdot 3.14) \\ &= 2114119.00 \text{ mm}^3\end{aligned}$$

Calculated Section Modulus (Zc):

$$\begin{aligned}&= I / Y \\ &= 20617/139 \\ &= 1485646.88 \text{ mm}^3 \text{ ...[Insufficient section modulus!]} \end{aligned}$$

Allowable Shear at the Pin Hole (Ssa):

$$\begin{aligned}&= 0.4 \cdot Syt \\ &= 0.4 \cdot 262 \\ &= 104.80 \text{ N./mm}^2\end{aligned}$$

Tail Lug Radius (R):

$$\begin{aligned}&= (TBA + HG + TTA) / 2 \\ &= (38 + 312 + 50) / 2 \\ &= 200.00 \text{ mm.}\end{aligned}$$

Calculated Shear at the Pin Hole (Ssc):

$$\begin{aligned}&= TLOAD / (2 \cdot TTL \cdot (R - DPH / 2)) \\ &= 38834 / (2 \cdot 40 \cdot (200 - 60/2)) \\ &= 28.00 \text{ N./mm}^2\end{aligned}$$

Allowable Bearing Stress at the Pin Hole (Sba):

$$\begin{aligned}&= 0.75 \cdot Syt \\ &= 0.75 \cdot 262 \\ &= 196.50 \text{ N./mm}^2\end{aligned}$$

Calculated Bearing Stress at the Pin Hole (Sbc):

$$\begin{aligned}&= TLOAD / (DPH \cdot TTL) \\ &= 38834 / (60 \cdot 40) \\ &= 158.68 \text{ N./mm}^2\end{aligned}$$

Weld Length (LW):

$$\begin{aligned}&= 2 \cdot (TOPWTH + TBA + TTA + HG + (DOU - DS) / 2) \\ &= 2 \cdot (216 + 38 + 50 + 312 + (5989 - 5557) / 2) \\ &= 1651.00 \text{ mm.}\end{aligned}$$

Allowable Weld Stress (Swa):

$$\begin{aligned}&= 0.49 \cdot Sat \\ &= 0.49 \cdot 138 \\ &= 67.62 \text{ N./mm}^2\end{aligned}$$

Calculated Weld Stress (Swc):

$$\begin{aligned}&= TLOAD / (0.707 \cdot WS \cdot LW) \\ &= 38834 / (0.707 \cdot 30 \cdot 1651) \\ &= 10.88 \text{ N./mm}^2\end{aligned}$$

Tailing Lug Results Summary (mm.³ & N./mm²):

	Actual	Allowable or Required	Pass/Fail
<i>Section Modulus</i>	<i>1485646.88</i>	<i>2114119.00</i>	<i>Failed</i>
Shear Stress	28.00	104.80	Passed
Bearing Stress	158.68	196.50	Passed
Weld Stress	10.88	67.62	Passed

[Review notes about nozzle loadings and supports in the Warnings report.](#)





Input, Nozzle Desc: N2-3"

From: 30

Pressure for Reinforcement Calculations	P	11.432	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	c	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	190.0000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		50.5500	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		3500.0	mm.
Centerline Distance to Tangent		1800.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N2-3"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 3.500 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 11.4)$
 $= 20.5444$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 11.4 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 11.4)$
 $= 0.4290$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 142.8618 mm.
 Parallel to Vessel Wall, opening length d 71.4309 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 49.8364 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	15.195	NA	NA
Area in Shell	A1	1.692	NA	NA
Area in Nozzle Wall	A2	7.079	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	15.109	NA	NA
TOTAL AREA AVAILABLE	Atot	24.735	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 108.7906 28.0000 mm.
 Based on given Pad Diameter: 190.0000 10.3209 mm.



Based on the Estimated Diameter Limit: 141.2743 10.6338 mm.

Area Required [A]:

$$\begin{aligned} &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\ &= (71.4 \cdot 20.5 \cdot 1 + 2 \cdot 8.73 \cdot 20.5 \cdot 1 \cdot (1 - 0.86)) \\ &= 15.195 \text{ cm}^2 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 71.4(1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 8.73 \\ &\quad (1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 0.86) \\ &= 1.692 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 49.8) \cdot (8.73 - 0.43) \cdot 0.86 \\ &= 7.079 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1 \\ &= 0.855 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (143 - 88.9) \cdot 28 \cdot 1 \\ &= 15.109 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.4290 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 23.8239 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 23.8239 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 5.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[5.8, \max(23.8, 3.5)] \\ &= 5.8000 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(1.43, 5.8) \\ &= 5.8000 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 36.3,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 258.7 N./mm ²	Passed
Occasional	: 2.1,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 24.1,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 0.43, c = 1 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.049, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.73, tr = 0.43, c = 1 mm., E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.049$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.73, tr = 0.43, c = 1 mm., E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.049$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 20.5, c = 1 mm., E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.89$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -8 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 0.43, c = 1 mm., E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.049$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $11.43/51.00 = 0.224$

Weld Size Calculations, Description: N2-3"

Intermediate Calc. for nozzle/shell Welds Tmin 8.7345 mm.
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 \cdot \text{TminPad}$	$14.1400 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (15.2 - 1.69 + 2 \cdot 8.73 \cdot 0.86 \cdot (1 \cdot 23 - 20.5)) 138)$
 $= 19517.26 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



$$\begin{aligned} &= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (7.08 + 15.1 + 0.86 - 0 \cdot 0.86) \cdot 138 \\ &= 32425.29 \text{ Kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (7.08 + 0 + 0.86 + (3.44)) \cdot 138 \\ &= 15998.63 \text{ Kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (7.08 + 0 + 0.86 + 15.1 + (3.44)) \cdot 138 \\ &= 37259.79 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 118 \\ &= 8233. \text{ Kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.14/2.0) \cdot 190 \cdot 20 \cdot 0.49 \cdot 138 \\ &= 41157. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 40.1) \cdot (9.73 - 1) \cdot 0.7 \cdot 118 \\ &= 9264. \text{ Kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.14/2) \cdot 88.9 \cdot 28 \cdot 0.74 \cdot 138 \\ &= 40716. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 88.9 \cdot (24 - 1) \cdot 0.74 \cdot 138 \\ &= 33444. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (41157 + 9264) = 50421 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (8233 + 40716 + 33444 + 0) = 82393 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (41157 + 33444 + 0) = 74601 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 50420	Kgf, must exceed W = 19517 Kgf	or W1 = 32425 Kgf
Path 2-2 = 82393	Kgf, must exceed W = 19517 Kgf	or W2 = 15998 Kgf
Path 3-3 = 74601	Kgf, must exceed W = 19517 Kgf	or W3 = 37259 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.1984 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.1985 mm.



Input Echo, WRC107/537 Item 1, Description: N2-3" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	9910.800	mm.
Vessel Thickness	Tv	24.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.0000	mm.
Nozzle Diameter	Dn	88.900	mm.
Nozzle Thickness	Tn	9.735	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	28.000	mm.
Diameter of Reinforcing Pad	Dpad	190.000	mm.
Design Internal Pressure	Dp	11.432	bars
Include Pressure Thrust		No	

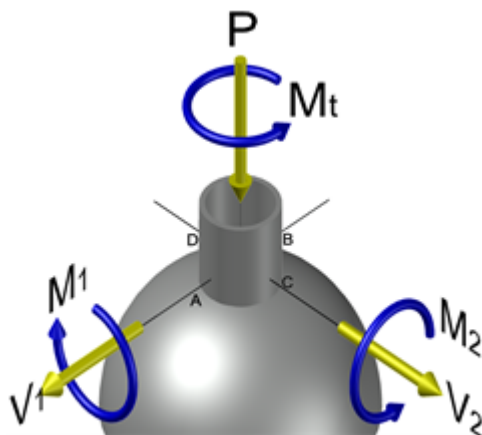
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	611.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	407.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	407.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	91.3	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	91.3	Kg-m.
Torsional Moment (SUS)	Mt	130.5	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Error/Warning Messages for WRC 107/537 Analysis: 1

Warning - The ratio R_o/R_m , 0.0089, must be > 0.0571 and < 0.571 .

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(4967.9 (24.0 - 1.0))}$$

$$= 1204.386 \text{ mm.}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	611.9	Kgf
Circumferential Shear	(VC) V2	407.9	Kgf
Longitudinal Shear	(VL) V1	407.9	Kgf
Circumferential Moment	(MC) M1	91.3	Kg-m.
Longitudinal Moment	(ML) M2	91.3	Kg-m.
Torsional Moment	MT	130.5	Kg-m.

$$\text{Unitless Prm: } U = 0.09 \text{ TAU} = 5.00 (4.59) \text{ RHO} = 4.00 (5.84)$$

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.11677	(A,B,C,D)
$M(x) / P$	SP 4	0.04430	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.60374	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.26038	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.60374	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.26038	(A,B,C,D)
$N(y) * T / P$	SP 4	0.32095	(A,B,C,D)
$M(y) / P$	SP 4	0.34978	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10351	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	2.22266	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10351	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	2.22266	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
Rad. Bend.	P	-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4



Rad. Bend. MC	0.0	0.0	0.0	0.0	-1.1	1.1	1.1	-1.1
Rad. Memb. ML	-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Rad. Bend. ML	-1.1	1.1	1.1	-1.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-2.4	1.0	0.6	-0.3	-2.4	1.0	0.6	-0.3
Tang. Memb. P	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Tang. Bend. P	-4.8	4.8	-4.8	4.8	-4.8	4.8	-4.8	4.8
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-9.1	9.1	9.1	-9.1
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-9.1	9.1	9.1	-9.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-14.8	13.1	3.6	-4.9	-14.8	13.1	3.6	-4.9
Shear VC	0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Tot. Shear	2.6	2.6	1.5	1.5	1.5	1.5	2.6	2.6
Str. Int.	15.3	13.7	4.2	5.5	14.9	13.3	6.0	6.9

Unitless Prm: U = 0.28 TAU = 0.00 (10.38) RHO = 0.00 (2.63)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.19757	(A,B,C,D)
M(x) / P	SR 2	0.14964	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16367	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.52097	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.16367	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.52097	(A,B,C,D)
N(y) * T / P	SR 2	0.05923	(A,B,C,D)
M(y) / P	SR 2	0.04492	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04990	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.15599	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04990	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.15599	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-2.2	-2.2	-2.2	-2.2	-2.2	-2.2	-2.2	-2.2
Rad. Bend. P		-10.2	10.2	-10.2	10.2	-10.2	10.2	-10.2	10.2
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-15.7	15.7	15.7	-15.7
Rad. Memb. ML		-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-15.7	15.7	15.7	-15.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-28.9	22.8	4.1	-6.9	-28.9	22.8	4.1	-6.9
Tang. Memb. P		-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Tang. Bend. P		-3.1	3.1	-3.1	3.1	-3.1	3.1	-3.1	3.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-4.7	4.7	4.7	-4.7
Tang. Memb. ML		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-4.7	4.7	4.7	-4.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-8.7	6.8	1.2	-2.1	-8.7	6.8	1.2	-2.1

Shear VC	0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.6	1.6	0.4	0.4	0.4	0.4	1.6	1.6
Str. Int.	29.0	22.9	4.1	6.9	28.9	22.8	4.7	7.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Rad.	Pl (SUS)	-0.7	-0.7	0.1	0.1	-0.7	-0.7	0.1	0.1
Rad.	Q (SUS)	-1.7	1.7	0.5	-0.5	-1.7	1.7	0.5	-0.5
Long.	Pm (SUS)	55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Long.	Pl (SUS)	-0.8	-0.8	-0.7	-0.7	-0.8	-0.8	-0.7	-0.7
Long.	Q (SUS)	-14.0	14.0	4.3	-4.3	-14.0	14.0	4.3	-4.3
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear	Q (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm (SUS)		55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Pm+Pl (SUS)		55.4	55.4	56.0	56.0	55.4	55.4	56.0	56.0
Pm+Pl+Q (Total)		53.7	69.2	59.7	55.7	53.4	68.9	60.6	56.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.56	138.00	Passed
Pm+P1 (SUS)	55.99	207.00	Passed
Pm+P1+Q (TOTAL)	69.23	413.99	Passed

*Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.*

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

[illegible]



Pm+Pl (SUS)	122.4	122.4	123.0	123.0	122.4	122.4	123.0	123.0
Pm+Pl+Q (Total)	114.6	146.1	127.3	121.2	114.5	146.0	127.9	121.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.19	138.00	Passed
Pm+Pl (SUS)	123.03	207.00	Passed
Pm+Pl+Q (TOTAL)	146.12	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K1B-2"

From: 40

Pressure for Reinforcement Calculations	P	11.432	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5506.00	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: K1B-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 11.4 \cdot 2754 / (138 \cdot 1 - 0.6 \cdot 11.4)$$

$$= 22.9295 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R (\exp([P/(S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.4 (\exp([11.4/(138 \cdot 1)] - 1))$$

$$= 0.2196 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	69.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12.107	NA	NA
Area in Shell	A1	3.481	NA	NA
Area in Nozzle Wall	A2	12.252	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	16.734	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (52.8 \cdot 22.9 \cdot 1 + 2 \cdot 15.8 \cdot 22.9 \cdot 1 \cdot (1 - 1))$$

$$= 12.107 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 85.5 (1 \cdot 27 - 1 \cdot 22.9) - 2 \cdot 15.8$$

$$(1 \cdot 27 - 1 \cdot 22.9) \cdot (1 - 1)$$

$$= 3.481 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 39.4) (15.8 - 0.22) 1$$

$$= 12.252 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.2196 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 3.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $tr_{b1} = 23.9295 \text{ mm.}$
 Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 23.9295 \text{ mm.}$
 Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 3.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[5.8, \max(23.9, 3.5)]$$

$$= 5.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.22, 5.8)$$

$$= 5.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 16.8$, $tr = 0.22$, $c = 1 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.014$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $7 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-104 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) $-104 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $11.43/51.00 = 0.224$

Weld Size Calculations, Description: K1B-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.7640 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (12.1 - 3.48 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 27 - 22.9)) \cdot 138)$$

$$= 13943.06 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (12.3 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 18647.77 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (12.3 + 0 + 1 + (8.51)) \cdot 138$$



= 30626.27 Kgf

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (12.3 + 0 + 1 + 0 + (8.51)) \cdot 138 \\ &= 30626.27 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138 \\ &= 9133. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 138 \\ &= 16724. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 84.3 \cdot (24 - 0) \cdot 0.74 \cdot 138 \\ &= 33104. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9133 + 16724) = 25857 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (9133 + 0 + 33104 + 0) = 42238 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (9133 + 33104 + 0) = 42238 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 25857	Kgf, must exceed W = 13943 Kgf	or W1 = 18647 Kgf
Path 2-2 = 42237	Kgf, must exceed W = 13943 Kgf	or W2 = 30626 Kgf
Path 3-3 = 42237	Kgf, must exceed W = 13943 Kgf	or W3 = 30626 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.3228 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.3229 mm.

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Input, Nozzle Desc: N6-2"

From: 40

Pressure for Reinforcement Calculations	P	11.432	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5506.00	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 11.4 \cdot 2754 / (138 \cdot 1 - 0.6 \cdot 11.4)$$

$$= 22.9295 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R (\exp([P/(S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.4 (\exp([11.4/(138 \cdot 1)] - 1))$$

$$= 0.2196 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	69.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12.107	NA	NA
Area in Shell	A1	3.481	NA	NA
Area in Nozzle Wall	A2	12.252	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	16.734	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (52.8 \cdot 22.9 \cdot 1 + 2 \cdot 15.8 \cdot 22.9 \cdot 1 \cdot (1 - 1))$$

$$= 12.107 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 85.5 (1 \cdot 27 - 1 \cdot 22.9) - 2 \cdot 15.8$$

$$(1 \cdot 27 - 1 \cdot 22.9) \cdot (1 - 1)$$

$$= 3.481 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 39.4) (15.8 - 0.22) 1$$

$$= 12.252 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$
$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.2196 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 23.9295 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 23.9295 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 5.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[5.8, \max(23.9, 3.5)]$$
$$= 5.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(1.22, 5.8)$$
$$= 5.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 11.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 333.2 N./mm ²	Passed
Occasional	: 0.7,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 9.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govern. thk, tg = 16.8, tr = 0.22, c = 1 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.014, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 11.43/51.00 = 0.224

Weld Size Calculations, Description: N6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) Sv)$$
$$= \max(0, (12.1 - 3.48 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 27 - 22.9)) 138)$$
$$= 13943.08 \text{ Kg}$$



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (12.3 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 18647.77 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (12.3 + 0 + 1 + (8.51)) \cdot 138$$

$$= 30626.27 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (12.3 + 0 + 1 + 0 + (8.51)) \cdot 138$$

$$= 30626.27 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9133. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 138$$

$$= 16724. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 84.3 \cdot (24 - 0) \cdot 0.74 \cdot 138$$

$$= 33104. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9133 + 16724) = 25857 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9133 + 0 + 33104 + 0) = 42238 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9133 + 33104 + 0) = 42238 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25857	Kgf, must exceed W = 13943 Kgf	or W1 = 18647 Kgf
Path 2-2 = 42237	Kgf, must exceed W = 13943 Kgf	or W2 = 30626 Kgf
Path 3-3 = 42237	Kgf, must exceed W = 13943 Kgf	or W3 = 30626 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.3228 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 234.6979 mm.

Input Echo, WRC297 Item 1, Description: N6-2" :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.0000 mm.
Shell External Corrosion Allowance	0.0000 mm.
Shell Diameter	5506.000 mm.



Shell Thickness		28.0000	mm.
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		1.0000	mm.
Nozzle Diameter		50.800	mm.
Nozzle Thickness		16.7640	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	11.43	bars
Radial Load	P	382.42	Kgf
Circumferential Shear	Vc	254.95	Kgf
Longitudinal Shear	Vl	254.95	Kgf
Circumferential Moment	Mc	35.68	Kg-m.
Longitudinal Moment	Ml	35.68	Kg-m.
Torsional Moment	Mt	50.97	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.012) is < 5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.030) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(2767.5(28.0 - 1.0))}$
 $= 986.397 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	5535.000	mm.
Nozzle Outside Diameter	(d) =	84.328	mm.
Vessel Thickness used	(T) =	27.000	mm.
Nozzle Thickness used	(t) =	15.764	mm.
T / t	=	1.713	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.218$$

Nr/P	=	0.267
Mr/P	=	0.315
M0/P	=	0.172
N0/P	=	0.180
MrD/Mc	=	0.250
NrDL/Mc	=	0.183
M0d/Mc	=	0.165
N0DL/Mc	=	-0.130
MrD/Ml	=	0.247

NrDL/M1 = 0.187
MØD/M1 = 0.163
NØDL/M1 = -0.116

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-5	5	-5	5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-5	5	5	-5
Normal Pressure Stress	116	117	116	117
Outplane Stress Summary	104	127	114	115

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-9	9	-9	9
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-8	8	8	-8
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	37	73	56	58

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	104	127	114	115

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-5	5	-5	5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-5	5	5	-5
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58
Outplane Stress Summary	46	68	56	56

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-9	9	-9	9
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-8	8	8	-8
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	117	116	117
Inplane Stress Summary	95	133	114	118

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	95	133	114	118

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	1	2	0	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-16	16	-16	16
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-15	15	15	-15
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-39	24	5	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary 4 4 1 1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress 41 25 6 8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3

Hoop Stress Summary 1 2 0 1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-16	16	-16	16
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-15	15	15	-15
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary -39 25 6 7

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	2	2	2	2

Shear Stress Summary 1 1 4 4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress 41 25 10 9

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	117	116	117	116	117	116	117
Circ. Pl (SUS)	0	0	0	0	-2	-2	0	0
Circ. Q (SUS)	-10	10	0	0	-17	17	-1	1
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-2	-2	0	0	0	0	0	0
Long. Q (SUS)	-17	17	-1	1	-10	10	0	0

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	116.0	117.0	116.0	117.0	116.0	117.0	116.0	117.0
Pm+P1 (SUS)	116.0	117.0	116.0	117.0	114.0	115.0	116.0	117.0
Pm+P1+Q (Total)	106.0	127.0	116.0	117.0	97.0	132.0	115.0	118.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	138.00	Passed
Pm+P1 (SUS)	117.00	207.00	Passed
Pm+P1+Q (TOTAL)	132.02	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	3	1	3	1	3	1	3
Circ. P1 (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. P1 (SUS)	-8	-8	6	6	-8	-8	6	6
Long. Q (SUS)	-31	31	-1	1	-31	31	-1	1
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0
Pm+P1 (SUS)	9.2	11.2	6.2	6.3	9.2	11.2	6.2	6.3
Pm+P1+Q (Total)	40.4	23.4	5.2	7.2	40.0	23.0	7.2	8.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+P1 (SUS)	11.18	207.00	Passed
Pm+P1+Q (TOTAL)	40.45	414.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: K1A-2"

From: 50

Pressure for Reinforcement Calculations	P	11.431	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5506.00	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: K1A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 11.4 \cdot 2754 / (138 \cdot 1 - 0.6 \cdot 11.4)$$

$$= 22.9289 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R (\exp([P/(S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.4 (\exp([11.4/(138 \cdot 1)] - 1))$$

$$= 0.2196 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	69.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12.106	NA	NA
Area in Shell	A1	3.482	NA	NA
Area in Nozzle Wall	A2	12.252	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	16.734	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (52.8 \cdot 22.9 \cdot 1 + 2 \cdot 15.8 \cdot 22.9 \cdot 1 \cdot (1 - 1))$$

$$= 12.106 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 85.5 (1 \cdot 27 - 1 \cdot 22.9) - 2 \cdot 15.8$$

$$(1 \cdot 27 - 1 \cdot 22.9) \cdot (1 - 1)$$

$$= 3.482 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 39.4) (15.8 - 0.22) 1$$

$$= 12.252 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.2196 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 23.9289 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 23.9289 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 5.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[5.8, \max(23.9, 3.5)]$$

$$= 5.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.22, 5.8)$$

$$= 5.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 16.8, tr = 0.22, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.014, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 11.43/51.00 = 0.224

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$$

$$= \max(0, (12.1 - 3.48 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 27 - 22.9)) \cdot 138)$$

$$= 13942.17 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (12.3 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 18647.77 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (12.3 + 0 + 1 + (8.51)) \cdot 138$$



= 30626.28 Kgf

Weld Load [W3]:

= $(A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
= $(12.3 + 0 + 1 + 0 + (8.51)) \cdot 138$
= 30626.28 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= $(\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
= $(3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$
= 9133. Kgf

Shear, Nozzle Wall [Snw]:

= $(\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
= $(3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 138$
= 16724. Kgf

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
= $(3.14/2.0) \cdot 84.3 \cdot (24 - 0) \cdot 0.74 \cdot 138$
= 33104. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= $(SONW + SNW) = (9133 + 16724) = 25857 \text{ Kgf}$

Failure Path [Path 2-2]:

= $(Sonw + Tpgw + Tngw + Sinw)$
= $(9133 + 0 + 33104 + 0) = 42238 \text{ Kgf}$

Failure Path [Path 3-3]:

= $(Sonw + Tngw + Sinw)$
= $(9133 + 33104 + 0) = 42238 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 25857 Kgf, must exceed W = 13942 Kgf	or W1 = 18647 Kgf
Path 2-2 = 42237 Kgf, must exceed W = 13942 Kgf	or W2 = 30626 Kgf
Path 3-3 = 42237 Kgf, must exceed W = 13942 Kgf	or W3 = 30626 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.3228 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.3229 mm.

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Input, Nozzle Desc: M1-24"

From: 60

Pressure for Reinforcement Calculations	P	11.431	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5506.00	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		7550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1009.6000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm.
Reinforcing Pad Width		200.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.024	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 11.4 \cdot 2754 / (138 \cdot 1 - 0.6 \cdot 11.4)$
 $= 22.9283 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 11.4 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 11.4)$
 $= 2.5166 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1119.2001 mm.
Parallel to Vessel Wall, opening length d 559.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 67.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	128.307	NA	NA
Area in Shell	A1	22.785	NA	NA
Area in Nozzle Wall	A2	30.353	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		5.000	NA	NA
Area in Element	A5	112.000	NA	NA
TOTAL AREA AVAILABLE	Atot	170.138	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 860.2029 28.0000 mm.
Based on given Pad Diameter: 1009.6000 17.5422 mm.
Based on Shell or Nozzle Thickness: 879.4801 26.0000 mm.

Area Required [A]:

$$\begin{aligned} &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\ &= (560 \cdot 22.9 \cdot 1 + 2 \cdot 25 \cdot 22.9 \cdot 1 \cdot (1 - 1)) \\ &= 128.307 \text{ cm}^2 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 560(1 \cdot 27 - 1 \cdot 22.9) - 2 \cdot 25 \\ &\quad (1 \cdot 27 - 1 \cdot 22.9) \cdot (1 - 1) \\ &= 22.785 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 67.5) \cdot (25 - 2.52) \cdot 1 \\ &= 30.353 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 + 20^2 \cdot 1 \\ &= 5.000 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (1010 - 610) \cdot 28 \cdot 1 \\ &= 112.000 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.5166 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 3.5166 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 26.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 26$, $t_r = 2.52$, $c = 1 \text{ mm.}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 26$, $t_r = 2.52$, $c = 1 \text{ mm.}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 26$, $t_r = 2.52$, $c = 1 \text{ mm.}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 28$, $t_r = 22.9$, $c = 1 \text{ mm.}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 26, tr = 2.52, c = 1 mm., E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -6 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -6 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $11.43/19.60 = 0.583$

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$13.3000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$14.1400 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (128 - 22.8 + 2 \cdot 25 \cdot 1 \cdot (1 \cdot 27 - 22.9)) \cdot 138)$
 $= 151349.89 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (30.4 + 112 + 5 - 0 \cdot 1) \cdot 138$
 $= 207348.17 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (30.4 + 0 + 1 + (13.5)) \cdot 138$
 $= 63114.55 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (30.4 + 0 + 5 + 112 + (13.5)) \cdot 138$
 $= 226344.80 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 66024. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 1010 \cdot 20 \cdot 0.49 \cdot 138$
 $= 218694. \text{ Kgf}$



Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 292) \cdot (26 - 1) \cdot 0.7 \cdot 138 \\ &= 226131. \text{ Kg} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.14/2) \cdot 610 \cdot 26 \cdot 0.74 \cdot 138 \\ &= 259246. \text{ Kg} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 610 \cdot (26 - 0) \cdot 0.74 \cdot 138 \\ &= 259246. \text{ Kg} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (218694 + 226131) = 444825 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (66024 + 259246 + 259246 + 0) = 584517 \text{ Kg} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (218694 + 259246 + 0) = 477941 \text{ Kg} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 444825 Kg	f, must exceed W = 151349 Kg	or W1 = 207348 Kg
Path 2-2 = 584516 Kg	f, must exceed W = 151349 Kg	or W2 = 63114 Kg
Path 3-3 = 477940 Kg	f, must exceed W = 151349 Kg	or W3 = 226344 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 16.9251 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 194.9252 mm.

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 4.455 \%$$

This nozzle or its diameter limit intersects weld seam # [4] on this element. The distance between the nozzle and seam is, 495.37 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 559.60 mm.

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Input, Nozzle Desc: N1-6"

From: 80

Pressure for Reinforcement Calculations	P	11.431	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5506.00	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	c	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		12000.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	270.0000	mm.
Thickness of Pad	te	26.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm.
Reinforcing Pad Width		50.8625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N1-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625	in.
Actual Thickness Used in Calculation	0.492	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 11.4 \cdot 2754 / (138 \cdot 1 - 0.6 \cdot 11.4)$
 $= 22.9274$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 11.4 \cdot 84.1 / (118 \cdot 1 + 0.4 \cdot 11.4)$
 $= 0.8119$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 290.5882 mm.
Parallel to Vessel Wall, opening length d 145.2941 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 54.7261 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	34.076	NA	NA
Area in Shell	A1	5.782	NA	NA
Area in Nozzle Wall	A2	9.994	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		3.913	NA	NA
Area in Element	A5	26.448	NA	NA
TOTAL AREA AVAILABLE	Atot	46.137	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 223.6092 26.0000 mm.
Based on given Pad Diameter: 270.0000 14.1429 mm.
Based on Shell or Nozzle Thickness: 283.4581 12.4904 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (145 \cdot 22.9 \cdot 1 + 2 \cdot 11.5 \cdot 22.9 \cdot 1 \cdot (1 - 0.86))$
 $= 34.076$ cm²

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 145 (1 \cdot 27 - 1 \cdot 22.9) - 2 \cdot 11.5$
 $(1 \cdot 27 - 1 \cdot 22.9) \cdot (1 - 0.86)$
 $= 5.782 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 54.7) \cdot (11.5 - 0.81) \cdot 0.86$
 $= 9.994 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$
 $= (1) \cdot 0.86 + (0) \cdot 0.86 + 306^2 \cdot 1$
 $= 3.913 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (270 - 168) \cdot 26 \cdot 1$
 $= 26.448 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.8119 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 23.9274 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 23.9274 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2200 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.22, \max(23.9, 3.5)]$
 $= 7.2200 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.81, 7.22)$
 $= 7.2200 \text{ mm.}$

Available Nozzle Neck Thickness = 12.4904 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	268.2 N./mm ²	Passed
Occasional	:	3.3,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	15.9,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 12.5$, $tr = 0.81$, $c = 1 \text{ mm.}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (t_g - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.5$, $tr = 0.81$, $c = 1 \text{ mm.}$, $E* = 1$
 Thickness Ratio = $tr \cdot E* / (t_g - c) = 0.071$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.5$, $t_r = 0.81$, $c = 1$ mm., $E^* = 1$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 26$, $c = 1$ mm., $E^* = 1$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.5$, $t_r = 0.81$, $c = 1$ mm., $E^* = 1$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -8 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -8 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $11.43/51.00 = 0.224$

Weld Size Calculations, Description: N1-6"

Intermediate Calc. for nozzle/shell Welds	T_{min}	11.4904 mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$14.1400 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (34.1 - 5.78 + 2 \cdot 11.5 \cdot 0.86 \cdot (1 \cdot 27 - 22.9)) \cdot 138)$
 $= 40940.36 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (9.99 + 26.4 + 3.91 - 0 \cdot 0.86) \cdot 138$
 $= 56786.79 \text{ Kg}$

Weld Load [W2]:



$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (9.99 + 0 + 0.86 + (5.31)) \cdot 138$$

$$= 22732.51 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (9.99 + 0 + 3.91 + 26.4 + (5.31)) \cdot 138$$

$$= 64252.72 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 15584. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 270 \cdot 20 \cdot 0.49 \cdot 138$$

$$= 58486. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 78.4) \cdot (12.5 - 1) \cdot 0.7 \cdot 118$$

$$= 23835. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 168 \cdot 26 \cdot 0.74 \cdot 138$$

$$= 71564. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 168 \cdot (26 - 0) \cdot 0.74 \cdot 138$$

$$= 71563. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (58486 + 23835) = 82321 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15584 + 71564 + 71563 + 0) = 158711 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (58486 + 71563 + 0) = 130049 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 82320 Kgf, must exceed W = 40940 Kgf	or W1 = 56786 Kgf
Path 2-2 = 158711 Kgf, must exceed W = 40940 Kgf	or W2 = 22732 Kgf
Path 3-3 = 130048 Kgf, must exceed W = 40940 Kgf	or W3 = 64252 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.2860 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.2860 mm.

Input Echo, WRC297 Item 1, Description: N1-6" :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.0000 mm.
Shell External Corrosion Allowance	0.0000 mm.

Shell Diameter		5506.000	mm.
Shell Thickness		28.0000	mm.
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		1.0000	mm.
Nozzle Diameter		168.275	mm.
Nozzle Thickness		12.4904	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad		270.000	mm.
Thickness of Reinforcing Pad		26.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	11.43	bars
Radial Load	P	1147.26	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(2780.5(28.0 - 1.0))}$$

$$= 1072.460 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	5561.000	mm.
Nozzle Outside Diameter	(d) =	168.275	mm.
Vessel Thickness used	(T) =	53.000	mm.
Nozzle Thickness used	(t) =	11.490	mm.
T / t	=	4.613	
d / t	=	14.645	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.310$$

Nr/P	=	0.169
Mr/P	=	0.124
M0/P	=	0.250
N0/P	=	0.283
MrD/Mc	=	0.125
NrDL/Mc	=	0.147
M0d/Mc	=	0.287
N0DL/Mc	=	-0.133
MrD/Ml	=	0.124

NrDL/M1 = 0.149
MØD/M1 = 0.279
NØDL/M1 = -0.087

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	58	59	58	59
Outplane Stress Summary	41	76	62	53

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-2	2	-2	2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	19	35	31	27

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	41	76	62	53

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29
Outplane Stress Summary	11	46	32	21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-2	2	-2	2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	59	58	59
Inplane Stress Summary	49	66	61	58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	49	66	61	58

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	9	8	9
Hoop Stress Summary	7	8	6	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-20	20	-20	20
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-15	-15	15	15
Axial Bending (ML)	-42	42	42	-42
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-75	49	39	-4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8



Shear Stress Summary 11 11 6 6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	85	52	40	17

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	9	8	9

Hoop Stress Summary 7 9 6 7

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-20	20	-20	20
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-43	43	43	-43
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary -76 50 40 -6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8

Shear Stress Summary 6 6 11 11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	85	51	43	26

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(2767.5(28.0 - 1.0))}$
 $= 1070.344 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	5535.000 mm.
Nozzle Outside Diameter	(d) =	270.000 mm.
Vessel Thickness used	(T) =	27.000 mm.
Nozzle Thickness used	(t) =	62.353 mm.
T / t	=	0.433



$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.698$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.123
Mr/P	=	0.277
M0/P	=	0.081
N0/P	=	0.150
MrD/Mc	=	0.312
NrDL/Mc	=	0.078
M0d/Mc	=	0.098
N0DL/Mc	=	0.022
MrD/M1	=	0.240
NrDL/M1	=	0.088
M0D/M1	=	0.077
N0DL/M1	=	0.087

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-7	7	-7	7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	116	117	116	117
Outplane Stress Summary	97	128	114	116

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-25	25	-25	25
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	6	103	54	60

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	97	128	114	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-7	7	-7	7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-9	9	9	-9
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58
Outplane Stress Summary	38	72	57	54

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-25	25	-25	25
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-29	29	29	-29
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	117	116	117
Inplane Stress Summary	57	169	119	112

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	57	169	119	112

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	59	58	59	58	59	58	59
Circ. Pl (SUS)	-1	-1	-1	-1	0	0	0	0
Circ. Q (SUS)	-17	17	5	-5	-6	6	2	-2
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	0	0	0	0	-1	-1	-1	-1
Long. Q (SUS)	-6	6	2	-2	-17	17	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	58.0	59.0	58.0	59.0	58.0	59.0	58.0	59.0
Pm+Pl (SUS)	57.0	58.0	57.0	58.0	58.0	59.0	58.0	59.0



Pm+Pl+Q (Total)	40.1	75.0	62.0	53.0	52.0	65.0	60.0	57.0
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	59.00	138.00	Passed
Pm+Pl (SUS)	59.00	207.00	Passed
Pm+Pl+Q (TOTAL)	75.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	117	116	117	116	117	116	117
Circ. Pl (SUS)	-3	-3	-1	-1	-2	-2	0	0
Circ. Q (SUS)	-14	14	0	0	-54	54	4	-4
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-2	-2	0	0	-2	-2	-2	-2
Long. Q (SUS)	-48	48	-2	2	-16	16	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	116.0	117.0	116.0	117.0	116.0	117.0	116.0	117.0
Pm+Pl (SUS)	113.0	114.0	115.0	116.0	114.0	115.0	116.0	117.0
Pm+Pl+Q (Total)	99.0	128.0	115.0	116.0	60.0	169.0	120.0	113.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	138.00	Passed
Pm+Pl (SUS)	117.00	207.00	Passed
Pm+Pl+Q (TOTAL)	169.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	9	8	9	8	9	8	9
Circ. Pl (SUS)	-1	-1	-1	-1	-1	-1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-16	-16	14	14	-16	-16	14	14
Long. Q (SUS)	-62	62	22	-22	-63	63	23	-23



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	8.0	9.0	8.0	9.0	8.0	9.0	8.0	9.0
Pm+Pl (SUS)	19.4	20.4	18.4	18.4	19.4	20.4	18.4	18.4
Pm+Pl+Q (Total)	83.4	52.3	41.1	17.0	82.9	51.8	43.7	23.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	118.00	Passed
Pm+Pl (SUS)	20.40	177.00	Passed
Pm+Pl+Q (TOTAL)	83.43	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N3-6"

From: 90

Pressure for Reinforcement Calculations	P	11.430	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	c	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	270.0000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		50.8625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N3-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625	in.
Actual Thickness Used in Calculation	0.492	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 11.4)$
 $= 20.5416 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 11.4 \cdot 84.1 / (118 \cdot 1 + 0.4 \cdot 11.4)$
 $= 0.8119 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 290.5882 mm.
 Parallel to Vessel Wall, opening length d 145.2941 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 56.7261 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	30.530	NA	NA
Area in Shell	A1	3.490	NA	NA
Area in Nozzle Wall	A2	10.359	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		3.501	NA	NA
Area in Element	A5	28.483	NA	NA
TOTAL AREA AVAILABLE	Atot	45.834	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 215.3430 28.0000 mm.
 Based on given Pad Diameter: 270.0000 12.9556 mm.
 Based on Shell or Nozzle Thickness: 273.7878 12.4904 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (145 \cdot 20.5 \cdot 1 + 2 \cdot 11.5 \cdot 20.5 \cdot 1 \cdot (1 - 0.86))$
 $= 30.530 \text{ cm}^2$



Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 145(1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 11.5$$

$$(1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 0.86)$$

$$= 3.490 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 56.7) \cdot (11.5 - 0.81) \cdot 0.86$$

$$= 10.359 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$

$$= (1) \cdot 0.86 + (0) \cdot 0.86 + 265^2 \cdot 1$$

$$= 3.501 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (270 - 168) \cdot 28 \cdot 1$$

$$= 28.483 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.8119 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 23.8209 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 23.8209 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.2200 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.22, \max(23.8, 3.5)]$$

$$= 7.2200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.81, 7.22)$$

$$= 7.2200 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.4904 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 268.2 N./mm ²	Passed
Occasional	: 3.3,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 15.9,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.5, tr = 0.81, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.071, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.5, tr = 0.81, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.071, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.5, tr = 0.81, c = 1 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 20.5, c = 1 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.89$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 12.5, tr = 0.81, c = 1 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $11.43/51.00 = 0.224$

Weld Size Calculations, Description: N3-6"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.4904 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (30.5 - 3.49 + 2 \cdot 11.5 \cdot 0.86 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$
 $= 38728.92 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (10.4 + 28.5 + 3.5 - 0 \cdot 0.86) \cdot 138$
 $= 59584.21 \text{ Kg}$



Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (10.4 + 0 + 0.86 + (4.52)) \cdot 138$
 $= 22140.43 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (10.4 + 0 + 3.5 + 28.5 + (4.52)) \cdot 138$
 $= 65944.08 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$
 $= 15584. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 270 \cdot 18 \cdot 0.49 \cdot 138$
 $= 52637. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 78.4) \cdot (12.5 - 1) \cdot 0.7 \cdot 118$
 $= 23835. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 168 \cdot 28 \cdot 0.74 \cdot 138$
 $= 77069. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 168 \cdot (24 - 1) \cdot 0.74 \cdot 138$
 $= 63306. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (52637 + 23835) = 76472 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (15584 + 77069 + 63306 + 0) = 155959 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (52637 + 63306 + 0) = 115943 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 76472 Kgf, must exceed W = 38728 Kgf	or W1 = 59584 Kgf
Path 2-2 = 155958 Kgf, must exceed W = 38728 Kgf	or W2 = 22140 Kgf
Path 3-3 = 115942 Kgf, must exceed W = 38728 Kgf	or W3 = 65944 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7116 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 424.7117 mm.

Input Echo, WRC107/537 Item 1, Description: N3-6" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical

Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	9910.800	mm.
Vessel Thickness	Tv	24.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	1.0000 mm.
Nozzle Diameter	Dn	168.275 mm.
Nozzle Thickness	Tn	12.490 mm.
Nozzle Material		SA-106 B
Nozzle UNS Number		K03006
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²
Thickness of Reinforcing Pad	Tpad	28.000 mm.
Diameter of Reinforcing Pad	Dpad	270.000 mm.
Design Internal Pressure	Dp	11.430 bars
Include Pressure Thrust		No

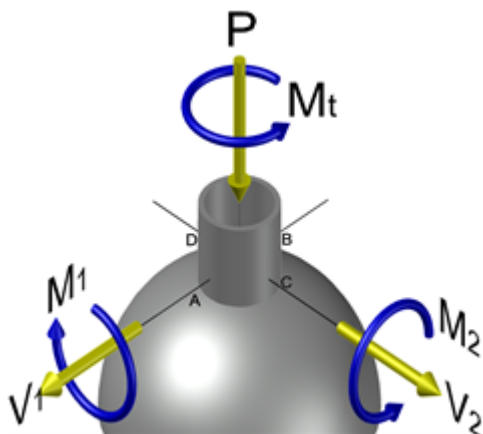
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	1147.3	Kgf
Longitudinal Shear (SUS)	(V1) V1	764.8	Kgf
Circumferential Shear (SUS)	(Vc) V2	764.8	Kgf
Circumferential Moment (SUS)	(Mc) M1	321.1	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	321.1	Kg-m.
Torsional Moment (SUS)	Mt	458.7	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(4967.9 (24.0 - 1.0))}$$

$$= 1283.761 \text{ mm.}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	1147.3	Kgf
Circumferential Shear	(VC) V2	764.8	Kgf
Longitudinal Shear	(VL) V1	764.8	Kgf
Circumferential Moment	(MC) M1	321.1	Kg-m.
Longitudinal Moment	(ML) M2	321.1	Kg-m.
Torsional Moment	MT	458.7	Kg-m.

Dimensionless Param: U = 0.17 TAU = 6.82 RHO = 4.00 (4.44)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.09126	(A,B,C,D)
M(x) / P	SP 4	0.03534	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.33604	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.14873	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.33604	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.14873	(A,B,C,D)
N(y) * T / P	SP 4	0.34493	(A,B,C,D)
M(y) / P	SP 4	0.22680	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.10275	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	1.06997	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.10275	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	1.06997	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4
Rad.	Bend. P	-0.9	0.9	-0.9	0.9	-0.9	0.9	-0.9	0.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-2.1	2.1	2.1	-2.1
Rad.	Memb. ML	-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-2.1	2.1	2.1	-2.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-4.3	1.9	1.6	-0.8	-4.3	1.9	1.6	-0.8
Tang.	Memb. P	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
Tang.	Bend. P	-5.9	5.9	-5.9	5.9	-5.9	5.9	-5.9	5.9
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-15.4	15.4	15.4	-15.4
Tang.	Memb. ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-15.4	15.4	15.4	-15.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-23.0	19.6	8.3	-10.8	-23.0	19.6	8.3	-10.8
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Tot. Shear		2.5	2.5	1.4	1.4	1.4	1.4	2.5	2.5
Str. Int.		23.4	19.9	8.6	11.0	23.2	19.7	9.1	11.4

Unitless Prm: U = 0.40 TAU = 0.00 (11.25) RHO = 0.00 (2.00)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.15984	(A,B,C,D)
M(x) / P	SR 2	0.10659	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.15697	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.33260	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.15697	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.33260	(A,B,C,D)
N(y) * T / P	SR 2	0.04802	(A,B,C,D)
M(y) / P	SR 2	0.03199	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04776	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.09985	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04776	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.09985	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-3.4	-3.4	-3.4	-3.4	-3.4	-3.4	-3.4	-3.4
Rad. Bend. P		-13.6	13.6	-13.6	13.6	-13.6	13.6	-13.6	13.6
Rad. Memb. MC		0.0	0.0	0.0	0.0	-2.8	-2.8	2.8	2.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-35.1	35.1	35.1	-35.1
Rad. Memb. ML		-2.8	-2.8	2.8	2.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-35.1	35.1	35.1	-35.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-54.9	42.6	20.9	-22.2	-54.9	42.6	20.9	-22.2
Tang. Memb. P		-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
Tang. Bend. P		-4.1	4.1	-4.1	4.1	-4.1	4.1	-4.1	4.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Tang. Bend. MC		0.0	0.0	0.0	0.0	-10.5	10.5	10.5	-10.5
Tang. Memb. ML		-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Tang. Bend. ML		-10.5	10.5	10.5	-10.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-16.5	12.8	6.3	-6.6	-16.5	12.8	6.3	-6.6
Shear VC		0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Tot. Shear		2.5	2.5	0.9	0.9	0.9	0.9	2.5	2.5
Str. Int.		55.1	42.8	21.0	22.2	54.9	42.6	21.3	22.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Rad. Pl (SUS)		-1.2	-1.2	0.4	0.4	-1.2	-1.2	0.4	0.4
Rad. Q (SUS)		-3.1	3.1	1.2	-1.2	-3.1	3.1	1.2	-1.2

Long. Pm (SUS)	55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Long. Pl (SUS)	-1.7	-1.7	-1.2	-1.2	-1.7	-1.7	-1.2	-1.2
Long. Q (SUS)	-21.3	21.3	9.5	-9.5	-21.3	21.3	9.5	-9.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm (SUS)	55.6	55.6	55.6	55.6	55.6	55.6	55.6	55.6
Pm+Pl (SUS)	54.7	54.7	56.1	56.1	54.7	54.7	56.1	56.1
Pm+Pl+Q (Total)	51.6	75.5	64.1	54.9	51.4	75.2	64.7	55.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.55	138.00	Passed
Pm+Pl (SUS)	56.13	207.00	Passed
Pm+Pl+Q (TOTAL)	75.47	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		123.2	123.2	123.2	123.2	123.2	123.2	123.2	123.2
Rad. Pl (SUS)		-6.2	-6.2	-0.6	-0.6	-6.2	-6.2	-0.6	-0.6
Rad. Q (SUS)		-48.7	48.7	21.5	-21.5	-48.7	48.7	21.5	-21.5
Long. Pm (SUS)		123.2	123.2	123.2	123.2	123.2	123.2	123.2	123.2
Long. Pl (SUS)		-1.9	-1.9	-0.2	-0.2	-1.9	-1.9	-0.2	-0.2
Long. Q (SUS)		-14.6	14.6	6.5	-6.5	-14.6	14.6	6.5	-6.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear Q (SUS)		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Pm (SUS)		123.2	123.2	123.2	123.2	123.2	123.2	123.2	123.2
Pm+Pl (SUS)		121.4	121.4	123.6	123.6	121.4	121.4	123.6	123.6
Pm+Pl+Q (Total)		106.8	166.0	144.1	116.6	106.7	165.8	144.5	116.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.17	138.00	Passed
Pm+Pl (SUS)	123.56	207.00	Passed
Pm+Pl+Q (TOTAL)	165.95	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N4-3"

From: 90

Pressure for Reinforcement Calculations	P	11.430	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	c	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	188.9000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		50.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N4-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 3.500 in.
Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 11.4)$
= 20.5416 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $11.4 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 11.4)$
= 0.4289 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 150.2074 mm.
Parallel to Vessel Wall, opening length d 75.1037 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 49.8364 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 118/138)
= 0.855

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 118/138)
= 0.855

Weld Strength Reduction Factor [fr4]:
= min(1, Sp/Sv)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.86, 1)
= 0.855

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	15.948	NA	NA
Area in Shell	A1	1.784	NA	NA
Area in Nozzle Wall	A2	7.412	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	15.886	NA	NA
TOTAL AREA AVAILABLE	Atot	25.937	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 72.01 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 109.9594 28.0000 mm.
Based on given Pad Diameter: 188.9000 10.3930 mm.
Based on the Estimated Diameter Limit: 148.6199 10.6922 mm.

Area Required [A]:



$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (75.1 \cdot 20.5 \cdot 1 + 2 \cdot 8.73 \cdot 20.5 \cdot 1 \cdot (1 - 0.86)) \\
 &= 15.948 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 75.1 (1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 8.73 \\
 &\quad (1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 0.86) \\
 &= 1.784 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3) \\
 &= (2 \cdot 49.8) \cdot (8.73 - 0.43) \cdot 0.86 / \sin(72.8) \\
 &= 7.412 \text{ cm}^2
 \end{aligned}$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1 \\
 &= 0.855 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (150 - 93.5) \cdot 28 \cdot 1 \\
 &= 15.886 \text{ cm}^2
 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 0.43, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.049, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.73, tr = 0.43, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.049, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.73, tr = 0.43, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.049, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 20.5, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.89, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 0.43, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.049, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 11.43/19.60 = 0.583

Weld Size Calculations, Description: N4-3"

Intermediate Calc. for nozzle/shell Welds Tmin 8.7345 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (15.9 - 1.78 + 2 \cdot 8.73 \cdot 0.86 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$
 $= 20446.90 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (7.41 + 15.9 + 0.86 - 0 \cdot 0.86) \cdot 138$
 $= 33987.04 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (7.41 + 0 + 0.86 + (3.44)) \cdot 138$
 $= 16467.19 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (7.41 + 0 + 0.86 + 15.9 + (3.44)) \cdot 138$
 $= 38821.54 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 93.5 \cdot 10 \cdot 0.49 \cdot 118$
 $= 8657. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 189 \cdot 18 \cdot 0.49 \cdot 138$
 $= 36827. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 42.1) \cdot (9.73 - 1) \cdot 0.7 \cdot 118$
 $= 9740. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:



$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.14/2) \cdot 93.5 \cdot 28 \cdot 0.74 \cdot 138 \\
 &= 42809. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.14/2.0) \cdot 93.5 \cdot (24 - 1) \cdot 0.74 \cdot 138 \\
 &= 35164. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (36827 + 9740) = 46567 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (8657 + 42809 + 35164 + 0) = 86630 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (36827 + 35164 + 0) = 71991 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 46567	Kgf, must exceed W = 20446 Kg	or W1 = 33987 Kg
Path 2-2 = 86629	Kgf, must exceed W = 20446 Kg	or W2 = 16467 Kg
Path 3-3 = 71990	Kgf, must exceed W = 20446 Kg	or W3 = 38821 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N4-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$\begin{aligned}
 &= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P) \text{ per Appendix 1-4(c)} \\
 &= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 \cdot 0.2 \cdot 11.4) \\
 &= 20.5416 \text{ mm.}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\
 &= 11.4 \cdot 44.5 / (118 \cdot 1 + 0.4 \cdot 11.4) \\
 &= 0.4289 \text{ mm.}
 \end{aligned}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	142.8618 mm.
Parallel to Vessel Wall, opening length	d	71.4309 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		49.8364 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	15.193	NA	NA
Area in Shell	A1	1.694	NA	NA



Area in Nozzle Wall	A2	7.079	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	15.109	NA	NA
TOTAL AREA AVAILABLE	Atot	24.737	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 108.7764 28.0000 mm.
 Based on given Pad Diameter: 188.9000 10.3136 mm.
 Based on the Estimated Diameter Limit: 141.2743 10.6262 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$
 $= (71.4 \cdot 20.5 \cdot 1 + 2 \cdot 8.73 \cdot 20.5 \cdot 1 \cdot (1 - 0.86))$
 $= 15.193 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 71.4(1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 8.73$
 $(1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 0.86)$
 $= 1.694 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 49.8) \cdot (8.73 - 0.43) \cdot 0.86$
 $= 7.079 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$
 $= 0.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (143 - 88.9) \cdot 28 \cdot 1$
 $= 15.109 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.4289 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 23.8209 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 23.8209 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(23.8, 3.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.43, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Weld Size Calculations, Description: N4-3"



Intermediate Calc. for nozzle/shell Welds T_{min} 8.7345 mm.
 Intermediate Calc. for pad/shell Welds T_{min}Pad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.
Pad Weld	9.5000 = 0.5*T _{min} Pad	12.7260 = 0.7 * W _p mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (15.2 - 1.69 + 2 \cdot 8.73 \cdot 0.86 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$
 $= 19512.32 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - \text{Can} / 707)^2 \cdot f_{r2}) \cdot S_v$
 $= (7.08 + 15.1 + 0.86 - 0 \cdot 0.86) \cdot 138$
 $= 32425.37 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (7.08 + 0 + 0.86 + (3.44)) \cdot 138$
 $= 15998.70 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (7.08 + 0 + 0.86 + 15.1 + (3.44)) \cdot 138$
 $= 37259.86 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 118$
 $= 8233. \text{ Kg}$

Shear, Pad Element Weld [S_{pew}]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 189 \cdot 18 \cdot 0.49 \cdot 138$
 $= 36827. \text{ Kg}$

Shear, Nozzle Wall [S_{nw}]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.1) \cdot (9.73 - 1) \cdot 0.7 \cdot 118$
 $= 9264. \text{ Kg}$

Tension, Pad Groove Weld [T_{pgw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 88.9 \cdot 28 \cdot 0.74 \cdot 138$
 $= 40716. \text{ Kg}$

Tension, Shell Groove Weld [T_{ngw}]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 88.9 \cdot (24 - 1) \cdot 0.74 \cdot 138$
 $= 33444. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (36827 + 9264) = 46091 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$



$$= (8233 + 40716 + 33444 + 0) = 82393 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (36827 + 33444 + 0) = 70271 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 46090 Kg, must exceed W = 19512 Kg or W1 = 32425 Kg

Path 2-2 = 82393 Kg, must exceed W = 19512 Kg or W2 = 15998 Kg

Path 3-3 = 70271 Kg, must exceed W = 19512 Kg or W3 = 37259 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.2871 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 439.4526 mm.

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Input, Nozzle Desc: N5-2"

From: 90

Pressure for Reinforcement Calculations	P	11.430	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	c	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1600.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 11.4)$
 $= 20.5416$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([11.4 / (138 \cdot 1)] - 1))$
 $= 0.2196$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 133.5780 mm.
 Parallel to Vessel Wall Rn+tn+t 66.7890 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.4100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	11.514	NA	NA
Area in Shell	A1	1.906	NA	NA
Area in Nozzle Wall	A2	12.956	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.862	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 70.39 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.1 \cdot 20.5 \cdot 1 + 2 \cdot 15.8 \cdot 20.5 \cdot 1 \cdot (1 - 1))$
 $= 11.514$ cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5(1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 1)$
 $= 1.906$ cm²

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 39.4)(15.8 - 0.22)/\sin(71)$
 $= 12.956$ cm²



See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govern. thk, $t_g = 16.8$, $t_r = 0.22$, $c = 1 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.014$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $7 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-104 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) $-104 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $11.43/51.00 = 0.224$

Weld Size Calculations, Description: N5-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 15.7640 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (11.5 - 1.91 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$
 $= 14610.13 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / 0.707)^2 \cdot fr_2) \cdot S_v$
 $= (13 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 19638.59 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$
 $= (13 + 0 + 1 + (7.25)) \cdot 138$
 $= 29842.51 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 $= (13 + 0 + 1 + 0 + (7.25)) \cdot 138$
 $= 29842.51 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 89.5 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9696. \text{ Kg}$

Shear, Nozzle Wall [Snw]:



$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 36.4) \cdot (16.8 - 1) \cdot 0.7 \cdot 138$$

$$= 17753. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 89.5 \cdot (24 - 1) \cdot 0.74 \cdot 138$$

$$= 33677. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9696 + 17753) = 27448 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9696 + 0 + 33677 + 0) = 43373 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9696 + 33677 + 0) = 43373 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 27448	Kgf, must exceed W = 14610 Kg	or W1 = 19638 Kg
Path 2-2 = 43372	Kgf, must exceed W = 14610 Kg	or W2 = 29842 Kg
Path 3-3 = 43372	Kgf, must exceed W = 14610 Kg	or W3 = 29842 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 \cdot 0.2 \cdot 11.4)$$

$$= 20.5416 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (Sn \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.4 \cdot (\exp([11.4 / (138 \cdot 1)] - 1))$$

$$= 0.2196 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	130.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	65.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100 mm.

Results of Nozzle Reinforcement Area Calculations: (cm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10.846	NA	NA
Area in Shell	A1	1.906	NA	NA
Area in Nozzle Wall	A2	12.252	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.158	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (52.8 \cdot 20.5 \cdot 1 + 2 \cdot 15.8 \cdot 20.5 \cdot 1 \cdot (1 - 1))$
 $= 10.846 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5(1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 1)$
 $= 1.906 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 39.4)(15.8 - 0.22)1$
 $= 12.252 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2196 \text{ mm}$.
Wall Thickness per UG16(b), $tr16b = 3.5000 \text{ mm}$.
Wall Thickness, shell/head, internal pressure $trb1 = 23.8209 \text{ mm}$.
Wall Thickness $tb1 = \max(trb1, tr16b) = 23.8209 \text{ mm}$.
Wall Thickness $tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm}$.
Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm}$.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(23.8, 3.5)]$
 $= 5.8000 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.22, 5.8)$
 $= 5.8000 \text{ mm}$.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: N5-2"

Intermediate Calc. for nozzle/shell Welds $Tmin = 15.7640 \text{ mm}$.

Results Per UW-16.1:

Nozzle Weld Required Thickness Actual Thickness
 $6.0000 = \text{Min per Code } 7.0700 = 0.7 \cdot Wo \text{ mm}$.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)



Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (10.8 - 1.91 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$
 $= 13670.71 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (12.3 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 18647.79 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (12.3 + 0 + 1 + (7.25)) \cdot 138$
 $= 28851.71 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (12.3 + 0 + 1 + 0 + (7.25)) \cdot 138$
 $= 28851.71 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9133. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 138$
 $= 16723. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 84.3 \cdot (24 - 1) \cdot 0.74 \cdot 138$
 $= 31724. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (9133 + 16723) = 25857 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (9133 + 0 + 31724 + 0) = 40858 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (9133 + 31724 + 0) = 40858 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 25856	Kgf, must exceed W = 13670 Kgf	or W1 = 18647 Kgf
Path 2-2 = 40857	Kgf, must exceed W = 13670 Kgf	or W2 = 28851 Kgf
Path 3-3 = 40857	Kgf, must exceed W = 13670 Kgf	or W3 = 28851 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.5336 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 439.8731 mm.

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Nozzle Calcs.: N5-2"



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Input, Nozzle Desc: K3-2"

From: 90

Pressure for Reinforcement Calculations	P	11.430	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	c	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1400.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 11.4)$
 $= 20.5416$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([11.4 / (138 \cdot 1)] - 1))$
 $= 0.2196$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 132.5677 mm.
 Parallel to Vessel Wall Rn+tn+t 66.2838 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.4100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	11.306	NA	NA
Area in Shell	A1	1.906	NA	NA
Area in Nozzle Wall	A2	12.739	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.644	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 73.60 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (55 \cdot 20.5 \cdot 1 + 2 \cdot 15.8 \cdot 20.5 \cdot 1 \cdot (1 - 1))$
 $= 11.306$ cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5(1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 1)$
 $= 1.906$ cm²

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 39.4)(15.8 - 0.22)/\sin(74.1)$
 $= 12.739$ cm²



See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govern. thk, $t_g = 16.8$, $t_r = 0.22$, $c = 1 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.014$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $7 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-104 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) $-104 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $11.43/51.00 = 0.224$

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 15.7640 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (11.3 - 1.91 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$
 $= 14318.08 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / 0.707)^2 \cdot fr_2) \cdot S_v$
 $= (12.7 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 19332.26 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$
 $= (12.7 + 0 + 1 + (7.25)) \cdot 138$
 $= 29536.17 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 $= (12.7 + 0 + 1 + 0 + (7.25)) \cdot 138$
 $= 29536.17 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 87.9 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9521. \text{ Kg}$

Shear, Nozzle Wall [Snw]:



$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 35.7) \cdot (16.8 - 1) \cdot 0.7 \cdot 138$$

$$= 17433. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 87.9 \cdot (24 - 1) \cdot 0.74 \cdot 138$$

$$= 33071. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9521 + 17433) = 26954 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9521 + 0 + 33071 + 0) = 42591 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9521 + 33071 + 0) = 42591 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 26953	Kgf, must exceed W = 14318 Kg	or W1 = 19332 Kg
Path 2-2 = 42591	Kgf, must exceed W = 14318 Kg	or W2 = 29536 Kg
Path 3-3 = 42591	Kgf, must exceed W = 14318 Kg	or W3 = 29536 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 \cdot 0.2 \cdot 11.4)$$

$$= 20.5416 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (Sn \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.4 \cdot (\exp([11.4 / (138 \cdot 1)] - 1))$$

$$= 0.2196 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	130.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	65.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100 mm.

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10.846	NA	NA
Area in Shell	A1	1.906	NA	NA
Area in Nozzle Wall	A2	12.252	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.158	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (52.8 \cdot 20.5 \cdot 1 + 2 \cdot 15.8 \cdot 20.5 \cdot 1 \cdot (1 - 1))$
 $= 10.846 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5 (1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 1)$
 $= 1.906 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 39.4) (15.8 - 0.22) \cdot 1$
 $= 12.252 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2196 \text{ mm}$.
 Wall Thickness per UG16(b), $tr16b = 3.5000 \text{ mm}$.
 Wall Thickness, shell/head, internal pressure $trb1 = 23.8209 \text{ mm}$.
 Wall Thickness $tb1 = \max(trb1, tr16b) = 23.8209 \text{ mm}$.
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm}$.
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm}$.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(23.8, 3.5)]$
 $= 5.8000 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.22, 5.8)$
 $= 5.8000 \text{ mm}$.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds $Tmin = 15.7640 \text{ mm}$.

Results Per UW-16.1:

Nozzle Weld Required Thickness Actual Thickness
 $6.0000 = \text{Min per Code } 7.0700 = 0.7 \cdot Wo \text{ mm}$.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)



Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (10.8 - 1.91 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$$

$$= 13670.69 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (12.3 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 18647.79 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (12.3 + 0 + 1 + (7.25)) \cdot 138$$

$$= 28851.71 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (12.3 + 0 + 1 + 0 + (7.25)) \cdot 138$$

$$= 28851.71 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9133. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 138$$

$$= 16724. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 84.3 \cdot (24 - 1) \cdot 0.74 \cdot 138$$

$$= 31725. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (9133 + 16724) = 25857 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (9133 + 0 + 31725 + 0) = 40858 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (9133 + 31725 + 0) = 40858 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25857	Kgf, must exceed W = 13670 Kgf	or W1 = 18647 Kgf
Path 2-2 = 40858	Kgf, must exceed W = 13670 Kgf	or W2 = 28851 Kgf
Path 3-3 = 40858	Kgf, must exceed W = 13670 Kgf	or W3 = 28851 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.5686 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 437.5744 mm.

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Nozzle Calcs.: K3-2"



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Input, Nozzle Desc: K2-2"

From: 90

Pressure for Reinforcement Calculations	P	11.430	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5506.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	c	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1600.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		250.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 11.4)$
 $= 20.5416$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([11.4 / (138 \cdot 1)] - 1))$
 $= 0.2196$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 133.5780 mm.
 Parallel to Vessel Wall Rn+tn+t 66.7890 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.4100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	11.514	NA	NA
Area in Shell	A1	1.906	NA	NA
Area in Nozzle Wall	A2	12.956	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.862	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 70.39 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.1 \cdot 20.5 \cdot 1 + 2 \cdot 15.8 \cdot 20.5 \cdot 1 \cdot (1 - 1))$
 $= 11.514$ cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5(1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 1)$
 $= 1.906$ cm²

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 39.4)(15.8 - 0.22)/\sin(71)$
 $= 12.956$ cm²



See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govern. thk, $t_g = 16.8$, $t_r = 0.22$, $c = 1 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.014$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $7 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-104 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) $-104 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $11.43 / 51.00 = 0.224$

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 15.7640 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (11.5 - 1.91 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$
 $= 14610.13 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / 0.707)^2 \cdot fr2) \cdot Sv$
 $= (13 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 19638.59 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (13 + 0 + 1 + (7.25)) \cdot 138$
 $= 29842.51 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (13 + 0 + 1 + 0 + (7.25)) \cdot 138$
 $= 29842.51 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 89.5 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9696. \text{ Kg}$

Shear, Nozzle Wall [Snw]:



$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 36.4) \cdot (16.8 - 1) \cdot 0.7 \cdot 138$$

$$= 17753. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 89.5 \cdot (24 - 1) \cdot 0.74 \cdot 138$$

$$= 33678. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9696 + 17753) = 27449 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9696 + 0 + 33678 + 0) = 43373 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9696 + 33678 + 0) = 43373 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 27448	Kgf, must exceed W = 14610 Kg	or W1 = 19638 Kg
Path 2-2 = 43373	Kgf, must exceed W = 14610 Kg	or W2 = 29842 Kg
Path 3-3 = 43373	Kgf, must exceed W = 14610 Kg	or W3 = 29842 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 11.4 \cdot 5508 \cdot 0.9 / (2 \cdot 138 \cdot 1 \cdot 0.2 \cdot 11.4)$$

$$= 20.5416 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (Sn \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.4 \cdot (\exp([11.4 / (138 \cdot 1)] - 1))$$

$$= 0.2196 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	130.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	65.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100 mm.

Results of Nozzle Reinforcement Area Calculations: (cm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10.846	NA	NA
Area in Shell	A1	1.906	NA	NA
Area in Nozzle Wall	A2	12.252	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.158	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (52.8 \cdot 20.5 \cdot 1 + 2 \cdot 15.8 \cdot 20.5 \cdot 1 \cdot (1 - 1))$
 $= 10.846 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5 (1 \cdot 23 - 1 \cdot 20.5) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 20.5) \cdot (1 - 1)$
 $= 1.906 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 39.4) (15.8 - 0.22) \cdot 1$
 $= 12.252 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.2196 \text{ mm.}$
Wall Thickness per UG16(b), $tr16b = 3.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure $trb1 = 23.8209 \text{ mm.}$
Wall Thickness $tb1 = \max(trb1, tr16b) = 23.8209 \text{ mm.}$
Wall Thickness $tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm.}$
Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(23.8, 3.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.22, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds $Tmin = 15.7640 \text{ mm.}$

Results Per UW-16.1:

Nozzle Weld Required Thickness Actual Thickness
 $6.0000 = \text{Min per Code } 7.0700 = 0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)



Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (10.8 - 1.91 + 2 \cdot 15.8 \cdot 1 \cdot (1 \cdot 23 - 20.5)) \cdot 138)$$

$$= 13670.71 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (12.3 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 18647.79 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (12.3 + 0 + 1 + (7.25)) \cdot 138$$

$$= 28851.71 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (12.3 + 0 + 1 + 0 + (7.25)) \cdot 138$$

$$= 28851.71 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9133. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 138$$

$$= 16724. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 84.3 \cdot (24 - 1) \cdot 0.74 \cdot 138$$

$$= 31725. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (9133 + 16724) = 25857 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (9133 + 0 + 31725 + 0) = 40858 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (9133 + 31725 + 0) = 40858 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25857	Kgf, must exceed W = 13670 Kgf	or W1 = 18647 Kgf
Path 2-2 = 40858	Kgf, must exceed W = 13670 Kgf	or W2 = 28851 Kgf
Path 3-3 = 40858	Kgf, must exceed W = 13670 Kgf	or W3 = 28851 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 11.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.5336 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 439.8731 mm.

Instrument Air reciever_
PV Elite 28 Licensee: SL User
FileName : Instrument Air reciever-7%
Nozzle Calcs.: K2-2"



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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: Tailing Lug Failure!**

Please review all reports carefully!

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